Product-Metric Certificates for Hierarchical Tucker Tree Selection

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Abstract

Choosing a hierarchical Tucker dimension tree for a PDE solution is difficult when the solution is not yet available. We study selection among prescribed rank-constrained families using the operator and forcing directly. The best squared energy error equals a common unknown constant minus an attainable variational gain, so feasible gains and upper bounds can certify a winning family without a full solution. For a Kronecker-sum operator, we construct a rank-preserving product metric whose centered spectral bound dominates the scalar-coercivity realization, even after center optimization. An explicit multiscale family has exact product bound one, whereas the best scalar bound remains of order $k^{2/3}$ even after optimizing its center, commuting product projection, and exact full-energy tail allowance. We also establish a spectral-tail enclosure and an unbounded gap for fully optimized all-edge forcing allocation with scalar operator splits. Gain intervals additionally bound the excess squared energy error of an actual feasible tensor, allowing termination without a unique winning tree. In controlled eight-mode experiments, guarded numerical intervals separate a winner among 315 balanced trees in all 14 non-tie cases using one feasible HT fit per case. Strong cached solve-and-round heuristics remain faster on these small problems. For affine forcing, support-function continuity transports certificates across parameter regions. A many-query study compares this reuse with offline basis solves, the same transport on solve-first certificates, and warm tree heuristics. Reuse reduces repeated certification work, but the solve-first reuse baseline remains faster on the tested sequences. The contribution is a structured comparison and certification framework, with conditional computational benefits.

1 Introduction

Hierarchical tensor formats represent a multivariate function through a tree of nested low-rank factorizations. The dimension tree determines which groups of variables must interact through small interfaces. Consequently, two trees with the same allocated ranks can differ substantially in approximation quality. For a supplied data tensor, one can inspect its matricizations or search its factorizations. For a solution defined implicitly by a partial differential equation (PDE), this strategy raises a circular dependency: the representation is needed to compute the solution efficiently, but the solution would ordinarily be used to choose the representation.

We consider a narrower, certifiable question. Given a finite-dimensional symmetric positive definite (SPD) problem and a collection of hierarchical Tucker (HT) families with prescribed ranks, can the best family in the operator's energy norm be identified before forming an accurate full solution? The forcing is part of the input. An operator alone cannot determine the answer for arbitrary right-hand sides: every candidate solution is realized by a suitable forcing under an invertible operator. We also distinguish selecting the best *family* from finding its globally best member. A useful family certificate need not require global convergence of the local optimizer used to construct a feasible approximation.

Our starting point is the standard variational energy identity. It converts best approximation error into a common solution-dependent constant minus a computable-objective optimum. The constant disappears when families are compared. The main challenge is then obtaining informative upper bounds on the energy each family can capture. A scalar coercivity bound provides one possibility, but suppresses the directional information of an elliptic operator. A general energy whitening transformation is not a remedy, because it need not preserve tensor ranks. We instead use positive product metrics that act separately on the two sides of an unfolding. These permit exact rank-restricted spectral calculations while retaining operator information.

The paper develops three connected results. First, an explicit balanced geometric metric for Kronecker sums dominates the scalar-coercivity centered bound pointwise in the center. This dominance survives exact optimization over centers. A multiscale diagonal example strengthens the comparison: the scalar/product ratio grows as $k^{2/3}$ even when the scalar method optimizes a commuting product projection and adds the exact full-energy tail. Second, rank-preserving spectral projection yields the tail enclosure used in that comparison. Third, we expose a distinct obstruction for multilevel HT: fully optimized forcing allocation across every edge, using scalar operator splits, can still have an unbounded gap relative to their intersection. Coupling cuts can tighten some comparisons but cannot remove that obstruction in general. For the compression example we also obtain the leading constant $3/2^{2/3}$ and prove the same order of separation for an ordinary two-dimensional Dirichlet difference operator with a prescribed spectral forcing. These are comparisons of relaxation strength, not runtime lower bounds or claims about typical loads.

We implement these bounds as a screening procedure: reuse shared cut calculations, build feasible approximations only for surviving families, and return a strict family certificate only when gain intervals separate. An additional stopping rule bounds the excess squared energy error of the computed tensor and can succeed at a tie. Our experiments separate three regimes: four-dimensional cut selection with nonseparable potentials, supplied spectral CP input with controlled omitted energy, and all-edge rank constraints on eight-dimensional HT trees. They also include direct spectral solves, preconditioned iterative solves, and an exhaustive rounding baseline that caches common subtrees. The strongest HT heuristic is faster on the reported small problems; the advantage there is an additional global family guarantee. This distinction is essential to the interpretation of the results.

For repeated problems with a fixed operator and affine forcing, we transport gain intervals using the support-function representation of their square roots. Homogeneous rescaling, intersections across anchors, and retained feasible directions provide sufficient parameter regions for reusing a certified family. We compare the complete sequence cost against a baseline that solves every forcing basis vector offline and then uses the same transport rule. This competitor remains faster in the reported experiment; the results distinguish the benefit of reuse from superiority of the direct seed certificate. The new repeated-problem experiments concern three four-dimensional cut families; the transport theorem itself also covers fixed all-edge HT families.

The multiscale comparison is a statement about a specified relaxation class: one retained product space, optimized scalar coercivity and center, and the unconstrained full-energy tail allowance. It does not compare with rank-aware tail estimates or allocation across spectral bands. The all-edge obstruction similarly concerns scalar operator splits, not every possible coupled relaxation.

The elementary variational identity, rank-restricted spectral conjugacy, and dual decomposition are established tools. The contribution studied here is their structured operator realization, the quantitative comparison and obstruction results, and an explicit implementation for tree-family certification. In particular, exact center optimization is a realization of known low-rank convexification after product whitening. We do not claim a general solution of tensor architecture search, adaptive rank allocation, or precedence over every existing preconditioned low-rank formulation.

2 Related work

Tensor structure selection. The hierarchical Tucker format represents a tensor through nested subspaces associated with a dimension tree; hierarchical singular value decomposition provides constructive approximation guarantees for a prescribed tree [1]. Automatic tree discovery is also established. Ballani and Grasedyck [2] infer a tree from selected tensor entries using agglomeration and rank-adaptive cross approximation. Grelier et al. [3] combine stochastic topology adaptation with rank selection in statistical learning, while Hikihara et al. [4] optimize tree tensor network connectivity through local reconnections in quantum variational calculations. More recently, Cadet separates dimension-tree enumeration from rank assignment and caches matricization spectra to assess many structures without decomposing every candidate [5]. HiST combines structural rounding, stochastic sub-network search, hierarchical refinement, and index reshaping [6]. These methods establish that tree search, local topology changes, and reuse of cut spectra are not new ingredients. Our target differs from compressing an available tensor in Frobenius norm: we compare the best energy errors of prescribed rank families for a solution specified implicitly by a linear equation. Applying a data-compression method to the forcing changes its approximation target. We therefore do not treat a poor

forcing-based tree as evidence against that method on its intended task, nor claim an empirical comparison with Cadet or HiST in the present experiments.

Variational low-rank PDE approximation. Computing a separated approximation directly from an elliptic variational problem is the purpose of proper generalized decomposition [7]. For positive definite Lyapunov equations, Vandereycken and Vandewalle [8] explicitly relate a computable quadratic objective to squared energy error up to an approximation-independent constant. The captured-energy identity used here is this standard variational principle, not a new way of eliminating an unknown solution. Alternating minimal energy methods combine local tensor optimization and residual-informed basis enrichment [9]; tensor Galerkin methods have established projection error analyses [10]. Ali and Nouy [11] analyze low-rank approximation in Sobolev norms and the error incurred by an SVD in the ambient L^2 norm. Bachmayr [12] surveys the broader theory of approximability, preconditioning, accuracy control, and computational complexity for tensor PDE solvers. Our question is a selection problem within this setting: can a feasible approximation in one family exclude every competing family through upper bounds on their attainable variational gains? Such a certificate does not require the feasible optimizer to reach its own family optimum. Conversely, removing full-grid solution iterations does not ensure a runtime advantage when the spectral bounds themselves are expensive. Our experiments count this work and include direct spectral solution whenever it is available.

Low-rank convexification and product metrics. The sum of the largest r squared singular values evaluates a rank-constrained quadratic conjugate. Explicit envelopes for fixed-rank approximation and the low-rank inducing norm framework are established [13, 14]. Our centered bound uses these results after rank-preserving left–right whitening. More precisely, optimizing its center gives the Fenchel dual of the convex problem that retains the $(\mathcal{A} - \mathcal{B})$ quadratic and replaces the rank-constrained $\mathcal{B}$ quadratic by its known convex envelope. Retaining a positive quadratic remainder is also part of quadratic-envelope regularization [15, Theorem 5.1]. Thus the construction is an operator-informed use of existing convexification. The balanced product minorant applies the arithmetic–geometric mean inequality after shifting the two spectral minima to coincide. Our quantitative comparison allows the scalar method to optimize its center, one retained product spectral space, and an exact omitted-energy allowance. It does not cover arbitrary rank-preserving preconditioning: a scalar method after the same product whitening realizes the product certificate itself.

Coupled unfolding bounds. The all-edge allocation norm is an instance of the duality between overlapped and latent Schatten norms [16, Lemma 1]: the relevant pair is the maximum unfolding nuclear norm and the infimum of sums of unfolding spectral norms. Our forcing allocation specializes this established principle to variational gain bounds. The matrix-multiplication tensor and its unit injective norm are known [17, Theorem 1.9 and Corollary 1.10]; see also Friedland et al. [18, Eq. (23)]. Combining its unfolding spectra with a common nuclear-norm witness gives the stated unbounded gap even after optimizing all forcing allocations and positive scalar operator splits. This identifies a limitation of that relaxation class; it does not exclude stronger couplings or general positive operator allocations.

Certified reuse and many-query computation. HiST already evaluates empirical transfer of optimized structures between related data instances [6, Figs. 12 and 14]. Our reuse criterion instead certifies that the same prescribed rank family remains best throughout a sufficient parameter region. Its square-root gain inequality is elementary support-function continuity, so the underlying perturbation principle is not a separate novelty claim. Safe screening provides a related exclusion methodology: Gap Safe constructs a ball containing a convex dual optimizer from a duality gap and combines screening with warm starts [19, Theorem 6 and Section 3.3]. Here the region lies in forcing space and preserves an ordering between nonconvex approximation families; it is not the Gap Safe dual-solution ball. The transport rule applies to any valid seed certificate, including solve-first certificates.

Offline–online reuse is standard in reduced-basis computation [20, Sections 1.3.4 and 1.3.6]. In particular, for fixed $\mathcal{A}$ and $f(\mu) = \sum_q \theta_q(\mu) f_q$, solving $\mathcal{A}u_q = f_q$ once gives $u(\mu) = \sum_q \theta_q(\mu) u_q$ exactly. Therefore avoiding a fresh solve at every parameter is not by itself an advantage over a strong baseline. Our many-query comparisons include these basis solves, reconstruction, warm tensor fits, and the same certificate

transport on competing valid seed bounds. The contribution sought is useful certification of family selection at a measured total cost, beyond empirical tree reuse or ordinary linear superposition.

3 Problem setting

Let $\mathcal{V} = \bigotimes_{i=1}^{d} \mathbb{R}^{n_i}$ have the Euclidean tensor inner product, and let $\mathcal{A} : \mathcal{V} \to \mathcal{V}$ be SPD. The unknown solution satisfies $\mathcal{A}u = f$. We write $\|x\|_{\mathcal{A}}^2 = \langle x, \mathcal{A}x\rangle$. Orthogonal changes of basis in each mode preserve this formulation and every matricization rank. All statements concern a fixed finite discretization; continuum discretization error is a separate issue.

A binary dimension tree partitions $\{1, \dots, d\}$ recursively. For each nonroot node t, let $x^{[t]}$ denote the unfolding with the modes below t as row indices. Given allocated ranks $\mathbf{r} = (r_t)$, define

$$\mathcal{K}_T = \{x \in \mathcal{V} : \operatorname{rank}(x^{[t]}) \le r_t \text{ for every nonroot node } t\}. \tag{1}$$

Singleton constraints are included. The root rank of the HT representation is one; r_t describes nonroot edges. Complementary children at the root give the same matricization rank. When equivalent cuts are identified, their effective rank cap is the minimum of the corresponding allocated caps. Each $\mathcal{K}_T$ is a closed cone containing zero and closed under arbitrary real scaling. The target is a minimizer over a finite, explicitly specified collection $\mathcal{T}$ of families of $\inf_{x \in \mathcal{K}_T} \|u - x\|_{\mathcal{A}}^2$. Equal rank budgets are imposed in our comparisons.

The main metric construction exploits the elliptic structure

$$\mathcal{A}_0 = \sum_{i=1}^{d} I \otimes \cdots \otimes A_i \otimes \cdots \otimes I, \qquad A_i \succ 0. \tag{2}$$

For a cut, this becomes $\mathcal{A}_0(X) = A_\ell X + X A_r$, with positive left and right cut operators. Four-dimensional experiments also consider $\mathcal{A} = \mathcal{A}_0 + \operatorname{diag}(V)$, $V \ge 0$, representing a nonnegative reaction potential on a Dirichlet finite-difference grid. Product minorants of $\mathcal{A}_0$ remain valid for $\mathcal{A}$; feasible energies and center corrections must still use the actual operator. The eight-dimensional implementation uses a separable spectral operator.

Define $g_{\mathcal{A},f}(x) = 2\langle f, x\rangle - \langle x, \mathcal{A}x\rangle$ and the attainable gain $G_T = \max_{x \in \mathcal{K}_T} g_{\mathcal{A},f}(x)$. The maximum exists by closedness and coercivity. The following sections construct lower and upper bounds on these gains using only the operator, forcing, and explicitly computed trial tensors.

4 Product-metric gain bounds

We work throughout with a fixed finite-dimensional real tensor-product space and a self-adjoint positive definite operator $\mathcal{A}$. All bounds in this section are exact-arithmetic statements. They compare approximation families for the discrete problem; they do not include PDE discretization error.

4.1 Comparing families through attainable energy

For an HT rank family $\mathcal{K}_T$, define

$$G_T = \sup_{x \in \mathcal{K}_T} g(x), \qquad g(x) = 2\langle f, x\rangle - \langle x, \mathcal{A}x\rangle. \tag{3}$$

Completing the square with $u = \mathcal{A}^{-1} f$ gives the standard Ritz identity

$$\inf_{x \in \mathcal{K}_T} \|u - x\|_{\mathcal{A}}^2 = \|u\|_{\mathcal{A}}^2 - G_T. \tag{4}$$

The unknown solution energy is common to every tree. Consequently, maximizing G_T selects the family with the smallest best energy approximation error, without requiring that common energy to be computed. This reformulation alone does not solve the resulting nonconvex optimization problem.

Because $\mathcal{K}_T$ is a cone closed under real scaling, every nonzero feasible tensor supplies a lower bound

$$\ell_T(x) = \frac{\langle f, x\rangle^2}{\langle x, \mathcal{A}x\rangle} \le G_T. \tag{5}$$

It is obtained by optimizing $g(tx)$ over $t \in \mathbb{R}$. If computable intervals satisfy $\ell_T \le G_T \le U_T$, then

$$\ell_{T_*} > \max_{T \neq T_*} U_T \tag{6}$$

certifies a unique best family among the specified candidates. Local tensor optimization can therefore contribute to a global selection certificate: its output needs to be feasible, rather than globally optimal.

Proposition 4.1 (Excess error of a computed tensor)**.** *Let $\widehat{x}$ be an explicitly computed tensor in one of the prescribed families, put $L = g(\widehat{x})$, and let $U = \max_T U_T$. Then*

$$0 \le \|u - \widehat{x}\|_{\mathcal{A}}^2 - \min_{T\in\mathcal{T}} \inf_{x\in\mathcal{K}_T} \|u - x\|_{\mathcal{A}}^2 = \max_T G_T - L \le U - L. \tag{7}$$

Thus $U - L \le \varepsilon$ certifies an absolute excess squared energy error of at most ε, even if several families have the same optimum.

Proof. Apply (4) to the computed tensor and the union of the candidate families, and subtract their common solution energy. Feasibility gives $L \le \max_T G_T$; the upper bounds give $\max_T G_T \le U$. □

This elementary consequence of the Ritz identity concerns the actual stored tensor. In particular U includes the selected family's own upper bound; winning the family comparison alone does not certify its local optimizer's accuracy. When using spectral compression, each U_T must include the full-space tail allowance. The quantity in (7) is not a relative error against the unknown optimal residual.

4.2 A centered spectral upper bound

First consider a single matricization, with forcing matrix F and feasible set $\{X : \operatorname{rank}(X) \le r\}$. Write its attainable gain as G_r. A product metric has the form $\mathcal{B}(X) = LXR$, where L, R are SPD matrices. Crucially, whitening by $L^{1/2}$ and $R^{1/2}$ preserves matrix rank. Define

$$\mathcal{H}_r(M) = \sum_{j=1}^{r} \sigma_j(M)^2. \tag{8}$$

This spectral conjugate is standard in low-rank convexification [14]; our use specializes it to operator-informed metrics.

Proposition 4.2 (Centered product-metric bound)**.** *Let $0 \prec \mathcal{B} \preceq \mathcal{A}$ and $\mathcal{C} = \mathcal{A} - \mathcal{B}$. For every matrix Y, of arbitrary rank,*

$$G_r \le \Phi_{\mathcal{B},r}(Y) := \langle Y, \mathcal{C}Y\rangle + \mathcal{H}_r\left(L^{-1/2}(F - \mathcal{C}Y)R^{-1/2}\right). \tag{9}$$

At $Y = 0$, the same truncated SVD yields the feasible direction

$$X_{\mathcal{B}} = L^{-1/2}\mathcal{P}_r\left(L^{-1/2}FR^{-1/2}\right)R^{-1/2}, \tag{10}$$

where $\mathcal{P}_r$ is any best rank-r Frobenius approximation. Evaluating its actual $\mathcal{A}$-energy gives the lower bound (5).

The proof drops the nonnegative term $\|X - Y\|_{\mathcal{C}}^2$ and evaluates a rank-constrained quadratic supremum by the SVD. Every center is valid, so one can retain the smallest bound observed without assuming that a center optimizer converged. The function $\Phi_{\mathcal{B},r}$ is convex in Y, but its optimized value can still exceed G_r. A direction satisfying one cut rank need not satisfy an entire HT tree; the latter requires a separate feasible tensor construction.

4.3 Dominance over scalar coercivity

Theorem 4.3 (Metric monotonicity and a balanced geometric metric)**.** *For positive product metrics $\mathcal{B}_1 \preceq \mathcal{B}_2 \preceq \mathcal{A}$, every common center satisfies*

$$G_r \le \Phi_{\mathcal{B}_2,r}(Y) \le \Phi_{\mathcal{B}_1,r}(Y). \tag{11}$$

In particular the same ordering holds after taking infima over all centers. For a Kronecker-sum operator $\mathcal{A}(X) = A_L X + X A_R$, let $a_0 = \lambda_{\min}(A_L)$, $b_0 = \lambda_{\min}(A_R)$, $m = a_0 + b_0$, and $s = (b_0 - a_0)/2$. Then

$$\mathcal{B}_*(X) = 2(A_L + sI)^{1/2} X (A_R - sI)^{1/2} \tag{12}$$

satisfies $mI \preceq \mathcal{B}_* \preceq \mathcal{A}$. Thus its centered bound is no larger than the bound from the largest scalar minorant mI, both pointwise and after exact center optimization.

The key identity behind monotonicity is the displacement representation

$$\Phi_{\mathcal{B},r}(Y) = g(Y) + \sup_{\operatorname{rank}(X)\le r} \left\{ 2\langle F - \mathcal{A}Y, X - Y\rangle - \|X - Y\|_{\mathcal{B}}^2 \right\}. \tag{13}$$

A larger minorant penalizes every displacement more strongly. The balanced metric shifts both factor spectra to the same minimum and applies the arithmetic–geometric mean inequality. If instead $\mathcal{A} \succeq \mathcal{A}_0$ for a separable minorant $\mathcal{A}_0$, its balanced metric remains valid, with the actual $\mathcal{A}$ used in the center correction. Dominance then concerns the scalar bound supplied by $\mathcal{A}_0$, not necessarily the strongest scalar bound for $\mathcal{A}$.

The advantage can persist even against a fully optimized scalar relaxation.

Theorem 4.4 (Unbounded optimized scalar/product gap)**.** *Fix $k \ge r \ge 1$ and $D > m > 0$, and set*

$$A_L = A_R = \operatorname{diag}(m/2, D/2, \dots, D/2), \qquad F = \operatorname{diag}(0, 1, \dots, 1), \tag{14}$$

with k high-mode entries. For $\mathcal{A}(X) = A_L X + X A_R$,

$$G_r = \Phi_{\mathcal{B}_*,r}(0) = \inf_Y \Phi_{\mathcal{B}_*,r}(Y) = \frac{r}{D}, \tag{15}$$

$$\inf_Y \Phi_{mI,r}(Y) = \frac{k}{D + m(k/r - 1)}. \tag{16}$$

The ratio of the optimized scalar bound to the product bound is $kD/[rD + m(k - r)]$. It is unbounded when r is fixed, $D/m = k^2$, and $k \to \infty$.

The example is an exact relaxation comparison, not a runtime comparison. Repeated high eigenvalues simplify its proof but are unnecessary for the phenomenon.

Corollary 4.5 (Four-dimensional Dirichlet example)**.** *Let n be even and let $\mathcal{A}$ be the standard four-dimensional Dirichlet finite-difference Laplacian on an n^4 grid. Its one-dimensional eigenvalues are $\lambda_p = 4(n+1)^2 \sin^2(p\pi/[2(n+1)])$. For the cut $12 \mid 34$, choose spectral forcing*

$$f = \sum_{p,q>n/2} \phi_p \otimes \phi_q \otimes \phi_p \otimes \phi_q. \tag{17}$$

For $1 \le r \le n^2/4$ and $m = 4\lambda_1$, the product bound is exact and

$$\frac{\inf_Y \Phi_{mI,r}(Y)}{G_r} \ge \frac{2n^2}{16r + \pi^2}. \tag{18}$$

These statements compare a particular structured realization with isotropic scalar coercivity. They do not claim dominance over arbitrary weighted low-rank convexifications, optimized product metrics, or other tree-search algorithms.

5 Compression and simultaneous tree ranks

5.1 A squared-tail allowance for spectral compression

A small retained tensor can suffice if its omitted forcing is controlled. Let $P = \bigotimes_{j=1}^{d} P_j$ be an orthogonal product projector commuting with $\mathcal{A}$, let $Q = I - P$, and set $h = Qf$. Such projectors arise by retaining selected eigenvectors of each axis operator in a Kronecker sum. Write $G_{T,P}$ for the gain restricted to $\mathcal{K}_T \cap \mathrm{range}(P)$.

Theorem 5.1 (One-sided spectral tail enclosure)**.** *For every prescribed HT rank family,*

$$G_{T,P} \le G_T \le G_{T,P} + E_{\mathrm{tail}}, \qquad E_{\mathrm{tail}} = \langle h, \mathcal{A}^{-1} h \rangle. \tag{19}$$

The full-space gain with forcing Pf equals $G_{T,P}$. Consequently, reduced intervals $\ell_{T,P} \le G_{T,P} \le U_{T,P}$ and any computable $E \ge E_{\mathrm{tail}}$ certify a full-space winner whenever

$$\ell_{T_*,P} > \max_{T \ne T_*} U_{T,P} + E. \tag{20}$$

The product structure ensures that Px preserves every HT rank constraint; commutation separates retained and omitted energies. We impose no rank condition on Qx. The allowance is quadratic in the omitted forcing, whereas a generic distance-to-proxy argument uses an allowance proportional to $\sqrt{E_{\mathrm{tail}}}$. These are different enclosures, not a universal runtime ordering. If λ_{out} lower-bounds the omitted operator spectrum, then $E_{\mathrm{tail}} \le \|h\|^2 / \lambda_{\mathrm{out}}$. CP-form input permits this norm to be computed from mode Gram matrices, without constructing the full tensor. Obtaining a CP representation of an arbitrary dense input is a separate cost.

5.2 A gap that survives spectral compression

The forcing in Theorem 4.4 is supported on a single high-frequency block. Restricting to that block improves the scalar coercivity constant and removes that example's scalar/product gap. We now compare the product bound with a stronger scalar procedure: choose any commuting orthogonal product projection, use the largest scalar minorant on its range, optimize the center there, and add the exact full-energy tail from Theorem 5.1. A different diagonal forcing leaves an unbounded gap even after all these choices are optimized.

We first give a formula for the scalar center optimization. This is a diagonal specialization of the spectral conjugate in (8); its role here is to make the stronger comparison explicit.

Proposition 5.2 (Diagonal scalar center optimization)**.** *Let $n \ge r \ge 1$, $d_i > 0$, $f_i \in \mathbb{R}$, and*

$$D = \mathrm{diag}(d_1, \dots, d_n), \qquad \mathcal{A}(X) = \tfrac{1}{2}(DX + XD), \qquad F = \mathrm{diag}(f_1, \dots, f_n).$$

For $0 < m \le \min_i d_i$, set $\Delta_{n,r} = \{p \in [0,1]^n : \sum_i p_i \le r\}$. Then

$$U_r(d, f; m) := \inf_Y \Phi_{mI,r}(Y) = \max_{p \in \Delta_{n,r}} \sum_{i=1}^{n} \frac{f_i^2 p_i}{m + (d_i - m)p_i}. \tag{21}$$

The formula includes coordinates with $d_i = m$. For fixed d, f, r, its value is nonincreasing in m on $(0, \min_i d_i]$.

The proof is given in Appendix A.9.

For the comparison below, let P_L, P_R be coordinate projectors associated with arbitrary subsets $L, R \subseteq \{1, \dots, k\}$, and write $\mathcal{P}_{L,R}(X) = P_L X P_R$. For $L, R \ne \varnothing$, the largest scalar minorant of $\mathcal{A}$ on this rectangular range is

$$m_{L,R} = \tfrac{1}{2}\left(\min_{i \in L} d_i + \min_{j \in R} d_j\right). \tag{22}$$

Let $U_{\mathrm{sc}}(L, R)$ denote the infimum over all centers of the rank-one scalar bound for the restricted operator, forcing $\mathcal{P}_{L,R}F$, and scalar metric $m_{L,R}I$. The exact tail is

$$E(L, R) = \langle F - \mathcal{P}_{L,R}F, \mathcal{A}^{-1}(F - \mathcal{P}_{L,R}F) \rangle. \tag{23}$$

If either set is empty, define $U_{\mathrm{sc}}(L,R) = 0$ and $E(L,R) = \langle F, \mathcal{A}^{-1}F\rangle$. The optimized scalar-plus-tail certificate is

$$V_k = \min_{L,R\subseteq\{1,\dots,k\}} \{U_{\mathrm{sc}}(L,R) + E(L,R)\}. \tag{24}$$

This comparator permits exact center optimization and exact tail energy; it uses a single retained product space and the full-energy tail allowance, without a rank constraint on the omitted component. It does not include a rank-dependent tail bound or joint allocation of a rank budget among multiple spectral bands.

Lemma 5.3 (Reduction to matched suffixes)**.** *Suppose $0 < d_1 < \cdots < d_k$ and $f_i = \sqrt{d_i}$. In (24), it suffices first to use matched sets $L = R = J$, and then to use suffixes $J_s = \{s+1,\dots,k\}$, $0 \le s \le k$. In particular,*

$$V_k = \min_{0\le s\le k} \{s + U_s\}, \qquad U_k = 0, \tag{25}$$

where, for $s < k$,

$$U_s = \max_{\substack{0\le p_i\le 1\ (i>s)\\ \sum_{i>s} p_i \le 1}} \sum_{i=s+1}^{k} \frac{d_i p_i}{d_{s+1} + (d_i - d_{s+1})p_i}. \tag{26}$$

The proof is given in Appendix A.10.

Theorem 5.4 (Unbounded gap after optimized spectral compression)**.** *Let $k \ge 64$ be even and set*

$$D_k = \mathrm{diag}(1^2, 2^2, \dots, k^2), \qquad A_L = A_R = \tfrac{1}{2}D_k, \qquad F_k = \mathrm{diag}(1,2,\dots,k). \tag{27}$$

For $\mathcal{A}(X) = A_L X + X A_R$ and the rank-one feasible set, the balanced product metric satisfies

$$G_1 = \Phi_{\mathcal{B}_*,1}(0) = \inf_Y \Phi_{\mathcal{B}_*,1}(Y) = 1. \tag{28}$$

In contrast, the optimized scalar-plus-tail certificate defined by (24) satisfies

$$\frac{1}{4}k^{2/3} \le V_k \le 3k^{2/3}. \tag{29}$$

The minimum in V_k includes every orthogonal product projector commuting with $\mathcal{A}$. The forcing has nonzero coefficients on every diagonal spectral mode, and $\kappa_2(\mathcal{A}) = k^2$.

The proof is given in Appendix A.11.

Each coordinate in both tensor factors is active in (27); this is full marginal spectral support, not a claim that every matrix coefficient is nonzero. In the orthonormal sine basis on $(0,\pi)^2$, the operator is the spectral Galerkin discretization of the scaled Dirichlet Laplacian $-\frac{1}{2}(\partial_x^2+\partial_y^2)$; the forcing in this construction depends on k. The result shows that the gap can survive optimal spectral compression with the tail allowance used here. It does not assert a gap against stronger rank-dependent tail estimates, spectral band allocations, or arbitrary weighted low-rank convexifications.

5.3 Edge bounds and an unavoidable relaxation gap

For a genuine HT tree, write $E(T)$ for its edge cuts, identifying complements, and impose all ranks $\mathrm{rank}(x^{(e)}) \le r_e$. Each individual rank family contains $\mathcal{K}_T$. Thus

$$G_T \le U_T^{\mathrm{edge}} := \min_{e\in E(T)} \min_{\mathcal{B},Y \text{ evaluated}} \Phi_{e,\mathcal{B},r_e}(Y). \tag{30}$$

A feasible HT tensor supplies the lower bound. Bounds for a shared cut can be cached across candidate trees, and a center can be reused regardless of its ranks. For eight binary modes at rank two, all 315 balanced labeled trees use only 63 distinct active cuts: 28 pairs and 35 complementary quartet pairs. Singleton constraints are then vacuous.

Proposition 5.5 (Independent edges can be arbitrarily loose)**.** *For each integer $n \ge 1$, there is a three-mode rank-one HT problem with $\mathcal{A} = I$ and $G_T = 1$ for which every exact single-edge relaxation equals n. Hence even exact operator metrics and arbitrary centers cannot uniformly bound the ratio U_T^{edge}/G_T.*

The construction uses the matrix-multiplication tensor and is given in Appendix A.7. It separates two obstructions: improving the operator metric cannot remove a gap caused by discarding simultaneous tree constraints.

5.4 Coupling two edges by forcing allocation

For two edges of the same tree, let $\mathcal{K}_1, \mathcal{K}_2$ denote their matrix-rank families. Fix $0 < \alpha < 1$ and an arbitrary tensor D, and split

$$f_1 = \alpha f + D, \quad f_2 = (1-\alpha)f - D, \qquad \mathcal{A}_1 = \alpha\mathcal{A}, \quad \mathcal{A}_2 = (1-\alpha)\mathcal{A}. \tag{31}$$

Let $\Phi_i(f_i; Y)$ be the centered product bound for cut i, operator $\mathcal{A}_i$, and any product metric $0 \prec \mathcal{B}_i \preceq \mathcal{A}_i$.

Proposition 5.6 (Joint allocation upper bound)**.** *For every allocation D and center Y,*

$$G_T \le J(D) := \Phi_1(\alpha f + D; Y) + \Phi_2((1-\alpha)f - D; Y). \tag{32}$$

The function J is convex. If each truncated SVD has a gap at its rank threshold, its gradient is $2(x_1(D) - x_2(D))$, where $x_i(D)$ is the corresponding weighted rank-constrained maximizer. At a tie, the same expression for any chosen maximizers is a subgradient.

This is a specialization of convex dual decomposition; allocation across tensor unfoldings has established antecedents [16]. Every evaluated allocation yields a valid upper bound, even if numerical optimization terminates early. We retain the minimum of joint and individual bounds: a fixed split and finite optimization budget need not improve the latter.

A strict improvement is already possible when there is no operator-metric error. For $\mathcal{A} = I$ and the binary tensor $f = e_{100} + e_{010} + e_{001}$, the exact independent rank-one cut bounds are 2. Couple the first two singleton cuts with the exact metrics $\mathcal{B}_1 = \alpha I$ and $\mathcal{B}_2 = (1-\alpha)I$, and write $J_\alpha(D)$ to display the split dependence. Then

$$G_T = \frac{4}{3} \;<\; \inf_{\substack{0<\alpha<1 \\ D}} J_\alpha(D) = \frac{25}{16} \;<\; 2. \tag{33}$$

The optimum is attained by $\alpha = 1/2$, $D_{100} = 1/8$, $D_{010} = -1/8$, with all other entries zero. A nuclear-norm witness proves its global optimality over every allocation and split (Appendix A.8). Thus two-edge coupling reduces the bound by 21.875%, but even its fully optimized value retains a strict gap.

5.5 An obstruction to allocating across all edges

The gap in Proposition 5.5 need not disappear when every available cut participates in the forcing allocation. The following statement concerns scalar operator splits $\mathcal{A}_e = \alpha_e I$ and their exact single-cut quadratic gains; it does not cover arbitrary positive-definite operator splits.

Proposition 5.7 (Unbounded gap for all-edge scalar allocation)**.** *Let $F \in (\mathbb{R}^{n^2})^{\otimes 3}$ be the matrix-multiplication tensor*

$$F_{(i,j),(j,k),(k,i)} = 1 \quad (1 \le i, j, k \le n), \qquad F_{a,b,c} = 0 \text{ otherwise.} \tag{34}$$

Take $\mathcal{A} = I$ and rank one on all three singleton cuts. For every $\alpha_1, \alpha_2, \alpha_3 > 0$ with $\sum_{e=1}^{3} \alpha_e = 1$, define

$$U_{\boldsymbol{\alpha}}(F) := \inf_{F_1+F_2+F_3=F} \sum_{e=1}^{3} \frac{\left\| F_e^{(e)} \right\|_{\mathrm{op}}^2}{\alpha_e}. \tag{35}$$

Then

$$G_T = 1, \qquad U_{\boldsymbol{\alpha}}(F) = n, \qquad \inf_{\substack{\alpha_e>0 \\ \sum_e \alpha_e = 1}} U_{\boldsymbol{\alpha}}(F) = n. \tag{36}$$

Thus the fully optimized all-edge allocation bound has an unbounded ratio to the attainable gain, even with exact single-cut metrics.

Proof. A tensor satisfying all three rank-one constraints is fully separable. Reshape its three unit factors into matrices X, Y, Z of unit Frobenius norm. The contraction with F is $\operatorname{tr}(XYZ)$, and

$$|\operatorname{tr}(XYZ)| \le \|XY\|_F \|Z\|_F \le \|X\|_F \|Y\|_F \|Z\|_F = 1.$$

Compatible matrix units attain equality. Optimizing the scalar multiplier of a unit separable tensor in $2\langle F, x\rangle - \|x\|_F^2$ therefore gives $G_T = 1$.

Every singleton unfolding of F has n^2 mutually orthogonal rows, each of squared norm n. Consequently,

$$\|F\|_F^2 = n^3, \qquad \left\|F^{(e)}\right\|_{\mathrm{op}}^2 = n, \qquad \left\|F^{(e)}\right\|_* = n^{5/2}, \quad e = 1, 2, 3,$$

where $\|\cdot\|_*$ denotes the matrix nuclear norm. The proportional allocation $F_e = \alpha_e F$ gives $U_{\boldsymbol{\alpha}}(F) \le \sum_e \alpha_e n = n$.

For the converse, spectral/nuclear norm duality and the scalar square inequality imply, for any tensor W and any allocation,

$$\frac{\left\|F_e^{(e)}\right\|_{\mathrm{op}}^2}{\alpha_e} \ge 2\langle F_e, W\rangle - \alpha_e \left\|W^{(e)}\right\|_*^2.$$

Choose the same witness $W = F/n^2$ for every cut. Then $\langle F, W\rangle = n$ and $\left\|W^{(e)}\right\|_*^2 = n$. Summing yields

$$\sum_{e=1}^{3} \frac{\left\|F_e^{(e)}\right\|_{\mathrm{op}}^2}{\alpha_e} \ge 2n - n\sum_{e=1}^{3} \alpha_e = n.$$

This holds for every allocation and every admissible weight vector, proving all asserted equalities. □

Remark 5.8 (Repeated scalar splits do not help). The same proof applies to any finite nonempty list of the three singleton cuts, including repetitions or complementary unfoldings, when each term has a positive scalar operator weight and the weights sum to one. The common witness gives the lower bound n, and proportional allocation attains it. In particular, adding further scalar splits of existing cuts cannot remove this obstruction. Since each split uses its exact metric $\alpha_e I$, unrestricted centers have no effect. The obstruction concerns the relaxation of simultaneous rank constraints, independently of center optimization or product-metric quality.

6 Screening tree families with gain intervals

We maintain $L_T \le G_T \le U_T$ for every candidate family. New feasible gains are intersected by taking a maximum; new upper bounds are intersected by taking a minimum. A family with $U_T < \max_S L_S$ cannot be uniquely optimal and needs no further fit. The general procedure is shown below. Its validity depends on the bounds and primal feasibility, not on the order of refinements or the convergence status of an optimization subroutine.

Algorithm 1: Variational tree-family screening

Input: $(\mathcal{A}, f)$, prescribed families $\mathcal{T}$, ranks, optional product projection, a finite refinement budget, and optionally an absolute excess squared energy-error tolerance ε.

1. Construct valid product metrics and cache upper bounds for distinct cuts. Include a certified spectral-tail allowance when using reduced input. Set $L_T = 0$ and U_T to the best available relaxation for each family.
2. Select a surviving family with largest upper bound. Construct a nested HT approximation, optionally improve its energy, and update L_T from its optimally rescaled feasible gain.
3. Refine upper bounds using unrestricted centers. When few competitors remain, try joint two-edge allocation bounds before constructing further primal candidates. Retain every previously obtained valid bound.
4. Let $S \in \arg\max_T L_T$. If $L_S > \max_{T \neq S} U_T$, return S with its gain intervals and positive separation margin. If an error tolerance was supplied and $\max_T U_T - L_S \leq \varepsilon$, return the stored tensor attaining L_S with the excess-error guarantee (7); this need not identify a unique tree.
5. Otherwise discard excluded competitors and continue while budget remains. Return an explicit abstention if intervals do not separate.

The implementation keeps these outcomes separate: `certified` denotes strict family separation, while `epsilon_certified` denotes the optional excess-error decision. The latter can be checked immediately after a fit, before paying for joint refinement. Its default tolerance is unset, preserving strict-selection behavior. On termination, the gain gap is reported in either mode; the actual optimally rescaled tensor can optionally be returned. Decisions include the numerical guard described in Appendix D and are not interval-arithmetic proofs.

Feasible multilevel candidates. Independent SVD truncations of unrelated unfoldings do not establish simultaneous HT feasibility. We construct nested frames bottom-up: leaf frames come from mode unfoldings, and an internal frame is selected inside the tensor product of its two child frames. Projection onto the two root frames then produces a tensor satisfying every allocated edge rank. The small core at each node can be updated by minimizing the actual quadratic energy while all other cores are fixed. These alternating updates preserve the representation dimensions. We evaluate the resulting energy before accepting a floating-point update and make no claim of global ALS optimality.

Initial operator-informed tensor. For (2), let $\mu_i = \lambda_{\min}(A_i)$, $\bar{\mu} = d^{-1} \sum_i \mu_i$, and $s_i = \bar{\mu} - \mu_i$. The modewise product

$$\mathcal{B}_{\mathrm{mode}} = d \bigotimes_{i=1}^{d} (A_i + s_i I)^{1/d} \preceq \mathcal{A}_0 \tag{37}$$

follows from arithmetic–geometric mean on the tensor-product eigenbasis. Its inverse acts modewise and preserves all unfolding ranks. We use $y_0 = \mathcal{B}_{\mathrm{mode}}^{-1} f$ as a common center and as the input to HT rounding. This is a product-preconditioner application, not the full inverse $\mathcal{A}_0^{-1} f$. The implementation can refine the center with a bounded number of preconditioned Richardson steps; their counts are reported separately. The final eight-mode study terminates before these optional steps in every non-tie case.

Shared work and joint refinement. Cuts are cached up to complementation. For eight binary modes and rank two, the 315 balanced trees use only 28 active pair cuts and 35 distinct root cuts; full-rank singleton constraints require no bound. Larger mode sizes also activate singleton constraints. Once at most six competing families remain, the implemented joint policy couples their root cut to up to two other restrictive cuts. It tries fixed operator fractions $\alpha \in \{0.5, 0.8, 0.95\}$ with at most 40 L-BFGS iterations per call, stopping as soon as an evaluated bound excludes the competitor. These fractions and the early-refinement schedule were chosen on development cases. At a singular-value tie the chosen spectral derivative is a subgradient; optimizer success is never used as a certificate.

Implementation scope. The 4D implementation combines small spectral spaces with full-space fallback bounds, including additive minorants of nonseparable potentials. The 8D implementation materializes a retained coefficient tensor and is intended to verify all-edge constraints and tree screening. It is not a compressed high-dimensional HT solver. Exact zero rows and columns may be removed from bound SVDs; no numerical threshold or primal support constraint is introduced by this operation. A separate CP-input route computes Gram matrices and omitted-energy bounds before assembling the reduced tensor. Detailed schedules and costs are given in the experimental appendix.

7 Reusing certificates for affine forcing

We now keep the operator, discretization, candidate families, and allocated ranks fixed, and vary only the forcing. Write $G_T(f) = \max_{x\in\mathcal{K}_T} g_{\mathcal{A},f}(x)$ and $s_T(f) = \sqrt{G_T(f)}$. The following support-function estimate provides a way to transport an already computed certificate. Its proof does not require the nonconvex set $\mathcal{K}_T$ to be convex.

Proposition 7.1 (Homogeneous transport of gain intervals)**.** *For each prescribed family,*

$$s_T(f) = \sup_{\substack{x\in\mathcal{K}_T\\ \|x\|_{\mathcal{A}}\le 1}} |\langle f, x\rangle|, \qquad |s_T(f) - s_T(h)| \le \|f-h\|_{\mathcal{A}^{-1}}\,. \tag{38}$$

Consequently, for every $t\in\mathbb{R}$,

$$\big|s_T(f) - |t| s_T(f_0)\big| \le \|f - tf_0\|_{\mathcal{A}^{-1}}\,. \tag{39}$$

If $0 \le L_{a,T} \le G_T(f_a) \le U_{a,T}$ at an anchor f_a, and $\delta_a \ge \|f - t_a f_a\|_{\mathcal{A}^{-1}}$, then

$$\Big(|t_a|\sqrt{L_{a,T}} - \delta_a\Big)_+ \le s_T(f) \le |t_a|\sqrt{U_{a,T}} + \delta_a. \tag{40}$$

Here $(z)_+ = \max\{z, 0\}$. The option $t_a = 1$ transports an ordinary neighborhood of an anchor. Allowing arbitrary real t_a also covers changes of amplitude and sign: exact collinearity $f = t_a f_a$ causes no transport loss. This homogeneity is relevant when forcing magnitude varies without changing its preferred family.

Computable parameter geometry. Let $f(\theta) = \sum_{j=0}^{p} c_j(\theta) f_j$, where $c(\theta) = (1, \theta_1, \dots, \theta_p)^{\mathsf{T}}$ for an affine parameterization. Choose a fixed minorant $0 \prec \mathcal{B} \preceq \mathcal{A}$ with accessible inverse action, and assemble

$$Q_{ij} = \langle f_i, \mathcal{B}^{-1} f_j\rangle, \qquad Q \succeq 0. \tag{41}$$

For anchor coefficients c_a, a valid radius is

$$\delta_a(t)^2 = (c - tc_a)^{\mathsf{T}} Q (c - tc_a) \ge \|f(\theta) - t f(\theta_a)\|^2_{\mathcal{A}^{-1}}\,. \tag{42}$$

If $c_a^{\mathsf{T}} Q c_a > 0$, the value

$$t_a^{\mathrm{proj}} = \frac{c_a^{\mathsf{T}} Q c}{c_a^{\mathsf{T}} Q c_a} \tag{43}$$

minimizes this radius over $t\in\mathbb{R}$. It need not maximize the eventual selection margin, because the transported anchor bounds also scale with $|t|$. Intersecting the bounds from $t = 1$ and $t = t_a^{\mathrm{proj}}$ is always valid. If $c_a^{\mathsf{T}} Q c_a = 0$, the anchor forcing is zero; we may take $t = 0$. No inverse of the possibly singular matrix Q is required.

For example, let $a = \sqrt{L_{a,S}}$ and $b = \max_{T\neq S}\sqrt{U_{a,T}}$, with $a > b$. With $t = 1$, the sufficient condition $2\delta_a(1) < a - b$ gives an open ellipsoid of strict family optimality in parameter space. It is an elliptic cylinder if the forcing directions are linearly dependent. For arbitrary t, the corresponding test is $2\delta_a(t) < |t|(a-b)$. Equality does not certify uniqueness.

A lower bound from a retained feasible direction. A stored nonzero tensor $x \in \mathcal{K}_T$ supplies a lower bound that can be reevaluated directly at every query. With

$$e_x = \langle x, \mathcal{A}x \rangle, \qquad v_{x,j} = \frac{\langle f_j, x \rangle}{\sqrt{e_x}}, \qquad \ell_x(c) = |c^{\mathsf{T}} v_x|, \tag{44}$$

we have $\ell_x(c) \le s_T(f(\theta))$. The squared lower bound is attained as a gain by the feasible tensor

$$\widehat{x}(c) = \frac{\langle f(\theta), x \rangle}{e_x} x, \qquad g_{\mathcal{A}, f(\theta)}(\widehat{x}(c)) = \ell_x(c)^2. \tag{45}$$

Reevaluation can improve on the transported anchor lower bound and does not require a new tensor optimization.

Corollary 7.2 (Intersected certificates and excess error)**.** *At a query, intersect any nonempty finite collection of valid transported intervals* (40)*, including different anchors, scales, and minorants. Let* $\underline{s}_T$ *be the maximum of their lower endpoints, zero, and every evaluated feasible-direction value* $\ell_x(c)$ *for that family; let* $\overline{s}_T$ *be the minimum of their upper endpoints. Then* $\underline{s}_T \le s_T(f(\theta)) \le \overline{s}_T$ *for every* T*. The strict test*

$$\underline{s}_S > \max_{T \ne S} \overline{s}_T \tag{46}$$

certifies a unique best prescribed family. If an actual returned feasible tensor has gain L_{ret}*, its excess squared energy error obeys*

$$\|u - \widehat{x}\|_{\mathcal{A}}^2 - \min_T \inf_{z \in \mathcal{K}_T} \|u - z\|_{\mathcal{A}}^2 \le \max_T \overline{s}_T^2 - L_{\mathrm{ret}}. \tag{47}$$

A transported lower bound concerns the best member of a family and need not be attained by a retained tensor at the new query. Therefore (47) uses the actual returned gain, for example (45), rather than an arbitrary transported lower endpoint. Failure of either test is an inconclusive reuse attempt, not evidence that the optimum changed. The ordinary selector may then refine the intervals and store an additional anchor; adaptive anchor selection does not change the validity of the pointwise inequalities.

Online work and a necessary baseline. For $q = p + 1$, the current implementation evaluates the quadratic forms separately for each of J anchors. Its query cost is $O(q^2 + J(q^2 + |\mathcal{T}|) + Rq)$ per dense metric and a fixed number of tested scales, where R is the number of retained feasible directions. Caching Qc_a and $c_a^{\mathsf{T}} Qc_a$ would reduce the exact-arithmetic operation count to $O(q^2 + Jq + J|\mathcal{T}| + Rq)$; this optimization is not implemented in the reported experiment. Multiple metrics add their corresponding costs. A single-anchor test could instead use a cached maximum competing upper endpoint, avoiding enumeration of all families; this shortcut need not be as strong as the full intersection. Gram assembly, inverse-metric applications, directional inner products, anchor certification, and tensor storage are offline costs and must be counted.

For fixed $\mathcal{A}$ and affine forcing there is also an exact offline solution-space construction:

$$u_j = \mathcal{A}^{-1} f_j \quad (0 \le j \le p), \qquad u(\theta) = \sum_{j=0}^{p} c_j(\theta) u_j. \tag{48}$$

Thus comparison only with a fresh PDE solve at every query would overstate the benefit of reuse. A fair computational comparison also permits these basis solves, reused tensor factorizations, and warm starts, and charges their preparation costs. The bounds above establish a certification mechanism; they do not by themselves establish an advantage in total runtime or in downstream solution accuracy. Proofs are in Appendix B.

8 Experiments

The experiments ask whether gain bounds can identify the energy-optimal *approximation family* before constructing a full solution, how much computation certification requires, and where the approach fails. We distinguish four-dimensional cut selection, compressed spectral input, and genuinely hierarchical rank constraints. The benchmarks are controlled finite-dimensional tests, not evidence of universal superiority over adaptive tensor solvers. Throughout, "certified" means that the implemented gain intervals separate after a floating-point guard; it does not mean an interval-arithmetic proof.

Group	n	Cases	Cert. ours	Cert. solve	R_{med}	R_{sum}
Development	8	18	18	18	4.14	1.75
Sensitivity + tie	8	21	19	15	1.82	0.22
Fresh seeds	8	18	18	18	4.04	1.71
Grid $n = 12$	12	3	3	3	15.03	2.22
Grid $n = 16$	16	3	3	3	32.90	1.95

Table 1: Four-dimensional selection. "Solve" is the fast complete solve followed by the residual-distance certificate. Each time is the median of three complete calls. With $t_{b,j}$ and $t_{o,j}$ denoting baseline and proposed times, $R_{\text{med}} = \text{median}_j(t_{b,j}/t_{o,j})$ and $R_{\text{sum}} = \sum_j t_{b,j} / \sum_j t_{o,j}$. Values above one favor our selector. Sensitivity contains 20 non-tie cases and one exact tie.

8.1 Four-dimensional energy selection

We discretize a Dirichlet elliptic operator on n^4 interior grid points, $A = \sum_{i=1}^{4} a_i L_i + \text{diag}(V)$, with $a_i > 0$ and $V \geq 0$. The three candidate families impose rank $r \in \{1, 2\}$ on the balanced cuts $12 \mid 34$, $13 \mid 24$, and $14 \mid 23$, respectively. Singleton ranks are unrestricted in this experiment; this is a matrix-rank comparison within a four-dimensional tensor, rather than the all-edge HT experiment below. Prescribed sources combine smooth pairwise interactions and a differently paired high-frequency component. Additional cases introduce anisotropic diffusion and a coupled, spatially varying potential. Source construction never uses a target solution.

Our selector starts with four retained spectral modes per axis and enriches its primal candidates and dual bounds only when needed. Difficult cases reach full-grid cut bounds, local constrained optimization, and unrestricted dual center refinement. No full-grid PDE iteration or reference solution enters this selector, but this statement does not exclude operator applications or constrained solves. Complete input processing and every fallback stage are included in its timings.

We compare against two certificate-producing implementations of the earlier residual-distance criterion: an online spectral-preconditioned CG proxy, and a fast complete solve followed by that criterion. Constant-potential problems use a direct spectral inverse; the remaining problems use spectral PCG. Both implementations cache operator factors and use optimized spectral-tail and leading-subspace computations. A third comparator solves first and chooses the best Frobenius cut of the solution. This inexpensive heuristic targets a different norm and provides no energy-family certificate.

Table 1 reports all groups, including abstentions. On the development and fresh-seed groups, all 18 comparisons are certified. Relative to solve-then-certify, median paired speed ratios are 4.14 and 4.04, while ratios of aggregate costs are 1.75 and 1.71. The distinction matters: two variable-potential, rank-two cases in each group are slower for our method. At $n = 16$, the paired-source and competing-source cases take about 6.1–6.2 ms versus 204 ms, whereas the variable-potential case takes 658 ms versus 900 ms. Thus the large median ratio of 32.9 does not describe the expensive potential case.

The sensitivity sweep is a substantive limitation. It increases successful certification from 15/20 to 19/20 non-tie cases, but its aggregate cost is 4.50 times that of solve-then-certify, despite a favorable median paired ratio of 1.82. The largest individual slowdown is 16.2 times. One near transition remains unresolved, and the exact tie correctly receives no strict certificate. Figure 1 retains these costly cases. The Frobenius heuristic is also cheaper in aggregate in every group; in two sensitivity cases its choice disagrees with our certified energy winner. The value of certification therefore depends on whether an energy guarantee is required.

8.2 Selection from supplied spectral CP factors

To separate dense input cost from bound evaluation, we supply an eleven-term spectral CP representation. It combines a low-frequency $12 \mid 34$ source, four times a high-frequency $13 \mid 24$ source, and three broad perturbation terms. Retaining indices $\{0, 1, 10, 11\}$ on each axis produces only $4^4 = 256$ entries; omitted modes are controlled by Gram-based tail bounds. For each $n \in \{16, 64, 256, 1024, 4096\}$, perturbation amplitudes 0, 0.1, and 0.5 yield certificates, while amplitude 1 yields an abstention. At $n = 4096$, seven-call median times range from 4.46 to 5.09 ms.

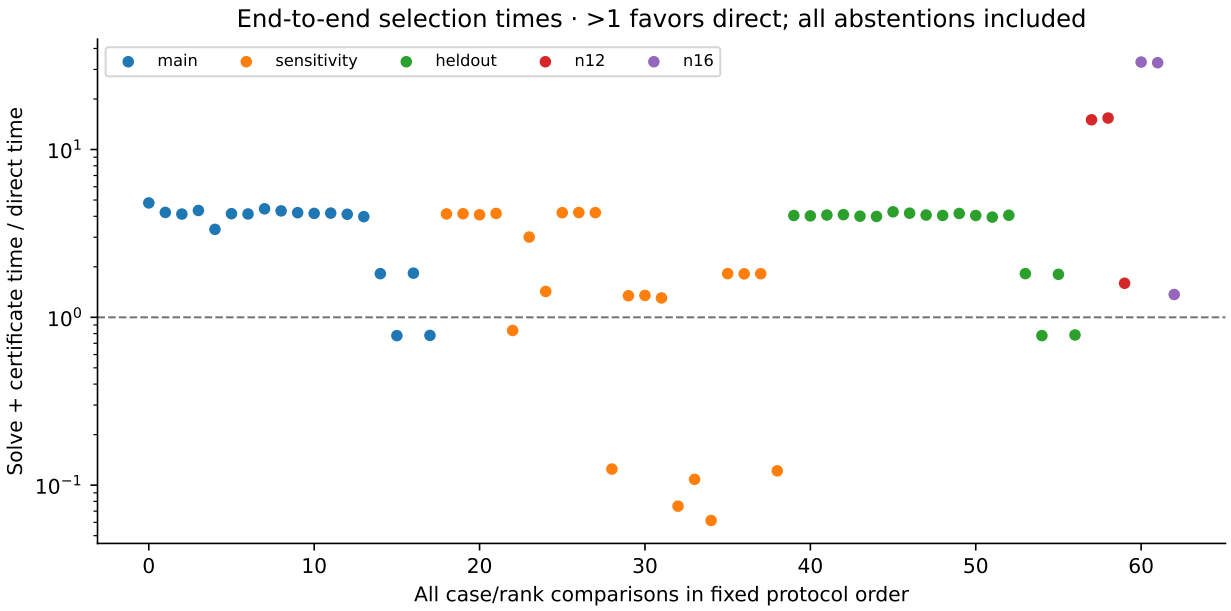


Figure 1: Every recorded four-dimensional case/rank comparison, in fixed protocol order. The ordinate is solve-then-certify time divided by our time, using within-case medians. Points below one are slowdowns. All abstentions and the exact tie are included; groups are diagnostic collections, not independent samples from a common population.

Eight-dimensional group	Cases	Cert.	Ours (ms)	Shared (ms)	R_{med}
Full modes, $n = 2$	3	3	74.0	39.7	0.54
Full modes, $n = 3$	3	3	274.9	101.5	0.37
Band development	5	5	729.9	539.8	0.78
Band fresh seeds	3	3	710.0	609.2	0.89
Exact tie	1	0	63.5	29.0	0.46

Table 2: Genuine HT selection among 315 balanced trees. Times are medians across within-case, three-repeat medians; R_{med} is the median paired shared-baseline/our-time ratio. Every proposed run uses one feasible HT fit and zero Richardson steps. The baseline is a heuristic, not a certificate.

These timings include compression, tail bounds, reduced assembly, and selection from already supplied factors. They exclude creating or learning that factorization. The largest input has 180,235 forcing-factor coefficients, in addition to 16,384 eigenvalues; no 4096^4 tensor is formed or solved. Dense $n = 16$ checks agree with every simultaneously certified choice. This is a compressed-input scaling demonstration, not a claim that arbitrary dense data can be processed at the same cost.

8.3 All-edge hierarchical Tucker selection

We next consider all 315 balanced binary dimension trees on eight labeled variables, imposing rank at most two on *every* nonroot edge, including singleton leaves. Sources are generated from random nested HT cores. Six small cases use full one-dimensional Dirichlet spectra with $n = 2$ or 3. Eight band cases mix differently structured low- and high-frequency sources on a 4^8 spectral tensor. An additional rank-one eigenfunction is an exact tie across all trees.

The selector certifies all 14 non-tie cases with one feasible HT fit and zero Richardson steps, and abstains on the tie. Joint-cut refinement is invoked only after single-cut bounds leave a short list. The comparator forms the exact spectral solution, rounds it into all 315 HT trees, ranks the candidates by actual rescaled energy gain, and applies two energy-ALS sweeps to the best. Identical leaf and subtree SVDs are shared across trees with fresh caches on each call. Its selected tree agrees with ours in every case.

Crucially, this strengthened exhaustive heuristic is faster on *every* non-tie case (Table 2). Reducing the number of feasible fits does not by itself establish a wall-time advantage: bounds and dual optimization have a cost. The demonstrated benefit is a certificate over the entire fixed family, obtained without forming the full solution. An ablation further shows that applying joint bounds early reduces HT fits from 12 to 1 and from 9 to 1 on two development cases, while eliminating ten proxy steps in each. Because those cases informed the schedule, three additional seeds were fixed before testing the final policy; all three are certified. These remain structured cases from the same generator, not an external validation distribution.

8.4 Optimized spectral compression in the gap example

We evaluate the diagonal family in Theorem 5.4 for $k \in \{16, 32, 64, 128, 256, 512, 1024, 2048, 4096\}$. The scalar comparator optimizes its unrestricted center and all retained suffixes, including the empty projection, with the exact omitted energy. By Lemma 5.3, this is also the optimum over all commuting product projections. Exhaustive enumeration of every matched subset for $k \leq 10$ independently verifies the suffix reduction in those small cases. A one-dimensional convex primal calculation and a separate simplex-dual calculation agree to at most 4.55×10^{-13} on the benchmark checks. This is a floating-point consistency check; the proof of the rate is analytical.

At $k = 4096$, the uncompressed optimized scalar bound is 4054.05. Optimal compression removes the lowest 261 modes and reduces its scalar plus exact-tail bound to 422.64, whereas the product bound and actual rank-one gain are both one. Figure 2 displays the comparison. These are relaxation values on a diagonal example, not runtime measurements or claims about general tree-search speed. The full table and reproduction commands are in Appendix D.6.

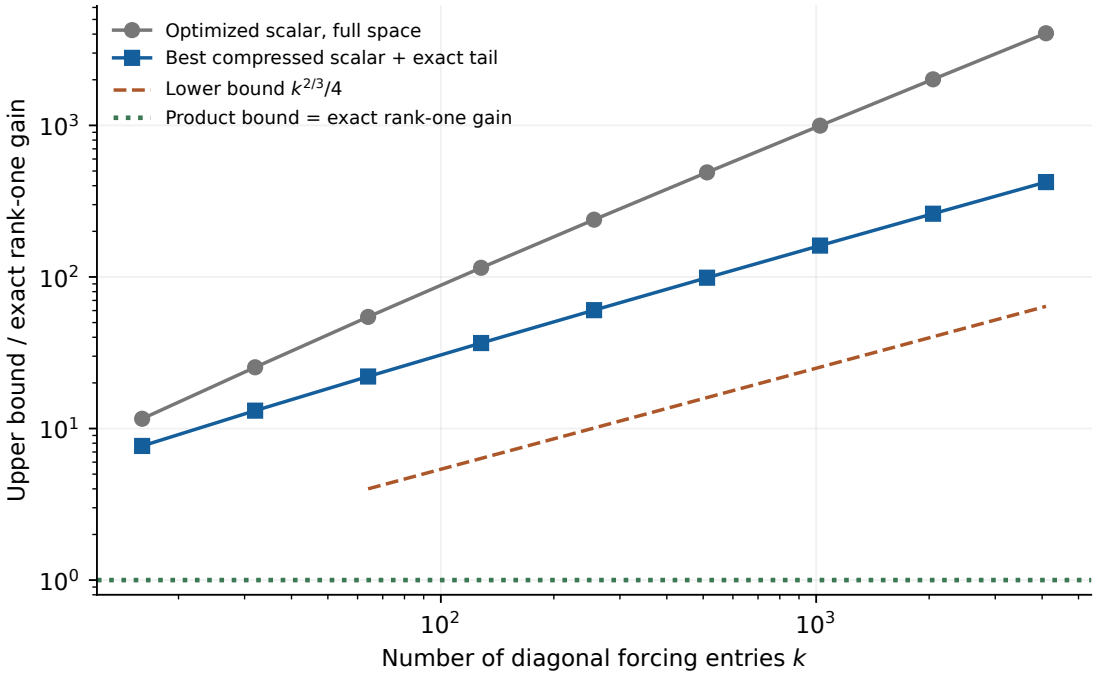


Figure 2: The scalar/product gap survives optimized spectral compression with the full-energy tail allowance. The theorem's lower bound is shown only for its stated range $k \geq 64$. The exact product value is one.

8.5 An excess-error stop at ties and near ties

A separate deterministic control uses $\mathcal{A} = I$ on four binary modes, rank at most two on every edge, and all three balanced trees. Put

$$a = \sum_{i,j\in\{0,1\}} e_i \otimes e_i \otimes e_j \otimes e_j, \qquad b = \sum_{i,j\in\{0,1\}} e_i \otimes e_j \otimes e_i \otimes e_j,$$

and normalize $f_\lambda = (1-\lambda)a + \lambda b$ to unit norm. The two leading singular values of each root unfolding give independent exact family gains because the leaf ranks are unrestricted at this mode size. These oracles are evaluated after the search, and directly check the squared energy error of its returned tensor.

The strict and optional excess-error policies use the same fitting and center budgets, with joint refinement disabled for this control. The latter uses absolute tolerance $\varepsilon = 10^{-8}$. At $\lambda = 0.5$ the strict policy correctly abstains after four fits, whereas the excess-error policy stops after one fit with a guarded excess bound below 10^{-9}. The same rule handles two perturbations 0.5 ± 10^{-12} that the numerical strict-separation guard leaves unresolved. At $\lambda = 0.49$ and 0.51, both policies stop after one fit. Table 3 reports deterministic work counts; it is not a runtime study.

Case	Strict fits	ε fits	Strict cert.	Excess bound
$\lambda = 0.49$	1	1	1	8.40×10^{-10}
$\lambda = 0.5 - 10^{-12}$	2	1	0	8.33×10^{-10}
$\lambda = 0.5$	4	1	0	8.33×10^{-10}
$\lambda = 0.5 + 10^{-12}$	2	1	0	8.33×10^{-10}
$\lambda = 0.51$	1	1	1	8.40×10^{-10}

Table 3: Tie controls at $\varepsilon = 10^{-8}$. Every excess-error run satisfies its tolerance and passes the independent SVD error oracle. "Strict cert." refers only to unique-family separation.

8.6 Repeated selection with a fixed reaction–diffusion operator

The preceding theory motivates a separate offline/online comparison with affine forcing. We use four coordinates, three balanced matrix cuts, and rank one on each candidate cut. These are the three 4D matrix-rank families, not the collection of 315 fully constrained eight-mode HT families. The operator is fixed within a sequence and has a nonseparable reaction potential. Six sequences use $n = 8$, seeds 41 and 67, three parameter paths, and 64 queries per path. Each method is rerun three times, with its execution order shuffled within each repetition; timings are medians over repetitions with one BLAS thread. A further three sequences use the held-out seed 89 on a 12^4 grid, 32 queries per path, and the same fixed decision policies and three timing repetitions.

Source and operator construction. Let $h = 1/(n+1)$ and use the standard centered Dirichlet difference operator on $(0,1)^4$. A seed fixes a coordinate permutation π and $\beta \in [8, 12]$. The diffusivities in the permuted coordinate order are $(1, 1.2, 0.8, 1.1)$. The potential is

$$V(x) = 5 + 35\left[1 + \tfrac{1}{4}\sin(2\pi(x_{\pi_1} + x_{\pi_3})) + \tfrac{1}{4}\cos(2\pi(x_{\pi_2} - x_{\pi_4}))\right]^2. \tag{49}$$

For the three distinct pairings $(a,b) \mid (c,d)$ of the permuted coordinates, the supplied forcing basis is

$$f_j(x) = h^2 \prod_{i=1}^{4} \sin(\pi x_i) \exp\left[-\beta\{(x_a - x_b)^2 + (x_c - x_d)^2\}\right]. \tag{50}$$

The factor h^2 gives the common quadrature scaling of the Euclidean coefficient tensors. The source shapes are specified directly and are not obtained from prescribed solution tensors. They remain analytic test loads, rather than measured application data.

At equally spaced $t \in [0,1]$, set $\alpha(t) = 1 + 0.4\sin(2\pi t)$ and $f(t) = \sum_{j=1}^{3} \theta_j(t) f_j$. The three paths are

$$\begin{aligned}
\theta_{\text{local}}(t) &= \alpha(t)(1,\ 0.025 + 0.01\sin(4\pi t),\ 0.025 + 0.01\cos(6\pi t)),\\
\theta_{\text{stable}}(t) &= \alpha(t)(1,\ 0.15 + 0.10\sin(4\pi t),\ 0.10 + 0.07\cos(6\pi t)),\\
\theta_{\text{transition}}(t) &= \alpha(t)(1-t,\ t,\ 0.03).
\end{aligned}$$

Thus the first two paths include shape changes as well as amplitude changes. The sampled transition paths each have one certified change of winning family. The reuse Gram matrix uses the separable minorant $\mathcal{B} = \mathcal{A}_0 + (\min_{\text{grid}} V)I$, whose inverse is applied spectrally. Both the unscaled and projected-scale transports are intersected, together with reevaluated feasible-direction lower bounds.

Six methods and charged work. Direct-fresh reruns the direct selector at every query; direct-reuse first attempts interval transport and invokes the same selector on a miss. The selector uses the compact metric policy, sweep caps $(0, 2, 6)$, and 12 dual refinements. Both share their operator setup across queries. Four solve-first competitors perform three offline basis solves with preconditioned CG at relative tolerance 10^{-11}, using a shared separable spectral preconditioner. Solve-certificate-fresh uses the residual-distance certificate at every reconstructed solution; solve-certificate-reuse applies the identical reuse mechanism to its gain intervals. This last method is essential: reuse is available to the solve-first certificate as well as to the direct certificate.

The solve-round-all heuristic evaluates all three rank-one rounded solutions at every query. The solve-warm-tree heuristic retains its previous family, refreshes all three cuts every 16 queries, and also refreshes when its monitored relative squared energy error exceeds 1.10 times its value at the last refresh. Both heuristics use the offline solution-energy information. Every method then fits the selected family with at most six alternating sweeps, uses a cached candidate as a warm start when available, and optimizes the final scalar amplitude. Failed warm fits and their subsequent refreshes are charged.

Table 4 reports complete sequence time, including operator and certificate setup, basis solves where used, Gram assembly, anchor construction, query checks, and every final fit. Only shared input generation and post-timing evaluation are excluded. Consequently these timings concern selection plus construction of an approximation, rather than the small parameter-space tests alone. Coverage and approximation diagnostics are reported in Table 5.

Grid/seed/path	N	D	D-R	S	S-R	H-all	H-warm
8/41/local	64	3.186	0.449	1.950	0.433	0.762	0.434
8/41/stable	64	3.463	0.884	1.972	0.538	0.906	0.498
8/41/transition	64	3.760	2.115	2.066	1.155	0.746	0.550
8/67/local	64	3.145	0.442	1.953	0.435	0.733	0.426
8/67/stable	64	3.390	1.760	2.091	0.547	0.780	0.607
8/67/transition	64	3.892	1.883	2.073	1.208	0.865	0.587
12/89/local	32	15.608	2.130	6.981	1.890	2.232	1.672
12/89/stable	32	15.267	7.305	6.583	2.344	2.167	1.822
12/89/transition	32	17.442	11.465	6.751	4.634	2.156	2.263

Table 4: Complete sequence time in seconds, median of three repetitions. D and D-R are fresh and reused direct certificates; S and S-R are fresh and reused solve-first certificates after three offline basis solves. H-all and H-warm are the all-cut and warm-tree solve-first heuristics. Each row includes its own setup and all returned-candidate fits. N is the number of queries.

Observed savings and the strongest competitor. For the six 8^4 sequences, direct-reuse certifies all 384 queries and handles 285 by transport, reducing fresh selections from 384 to 99. Its sum of median sequence times is 7.534 s, compared with 20.836 s for direct-fresh: an aggregate factor 2.77, with individual sequence factors from 1.78 to 7.12. However, solve-certificate-reuse is faster than direct-reuse in every sequence. Its aggregate time is 4.316 s, with 320 reuse hits and 380 certified queries; the four unresolved queries lie next to the two observed family transitions. Relative to solve-certificate-fresh, it improves aggregate time from

Grid/seed/path	Cert. D-R	Cert. S-R	Hits D-R	Hits S-R	E^2_{D-R} (%)	E^2_{H-warm} (%)
8/41/local	64/64	64/64	63	63	0.843	0.843
8/41/stable	64/64	64/64	54	60	0.910	0.910
8/41/transition	64/64	62/64	32	36	2.618	2.674
8/67/local	64/64	64/64	63	63	0.826	0.826
8/67/stable	64/64	64/64	36	60	0.936	0.936
8/67/transition	64/64	62/64	37	38	3.224	3.325
12/89/local	32/32	32/32	31	31	0.806	0.806
12/89/stable	32/32	32/32	19	28	0.940	0.940
12/89/transition	32/32	30/32	12	14	3.562	3.562

Table 5: Strict certificate coverage, reuse hits, and maximum relative squared energy error in percent over each complete sequence. D-R is direct reuse and S-R is solve-first reuse. H-warm supplies no strict certificate. Both solve-first certificate variants and H-all attain the same observed maximum error as D-R to the displayed precision. Counts refer to distinct queries, not timing repetitions.

12.106 s by a factor 2.80. Thus reuse benefits both certificate constructions; these tests do not establish a general runtime advantage of direct certification over an affine solve-first strategy.

The held-out 12^4 sequences show the same limitation. Direct-reuse certifies all 96 queries with 62 reuse hits and reduces aggregate time from 48.317 s to 20.901 s, a factor 2.31. Individual path factors range from 1.52 to 7.33. Solve-certificate-reuse is again faster on every path, taking 8.868 s with 73 hits and 94 certified queries; its two unresolved queries bracket the sampled family transition. All methods have the same observed maximum relative squared energy error, 0.03562, and no stale-family decision is observed for the warm-tree heuristic on this coarser query sampling. It takes 5.757 s in aggregate. These held-out paths change both seed and grid size, so this comparison is not a controlled grid-convergence study.

Returned approximation quality. Post-timing basis solves at tolerance 10^{-12} supply solution-energy diagnostics. We report $\|u - \widehat{x}\|^2_{\mathcal{A}} / \|u\|^2_{\mathcal{A}}$, the relative *squared* energy error, using the midpoint of the residual-based solution-energy interval. For the 8^4 tests its maximum is 0.03224 for direct-reuse and the solve-certificate and solve-round-all competitors. Their observed gains agree to numerical precision. Solve-warm-tree is faster in aggregate (3.103 s), but returns the previous family for one query after each transition. Its largest excess over the best observed gain, divided by the solution energy, is 5.58×10^{-4} for seed 41 and 1.00×10^{-3} for seed 67; its maximum total relative squared energy error is 0.03325. The resulting loss is small but measurable.

Successful family transport does not imply an equally tight error certificate for the computed tensor. Although the final fits closely match the best observed gains, the reused interval can remain loose: the largest reported excess-error upper bound for direct-reuse is 0.08734 times the solution energy on the 8^4 cases. Hence the strict family decisions and the quality of the returned tensor are assessed separately. No conflicting strict decisions, upper bounds excluding an independently computed feasible candidate, or repeat inconsistencies were observed. These checks are numerical evidence and do not convert the floating-point implementation into interval arithmetic.

9 Discussion and limitations

The results separate two sources of relaxation error. An isotropic penalty can discard essential operator geometry even after its center and retained spectral product space are optimized, when the omitted component is covered by a full-energy tail. A product metric is exact on the multiscale example where that scalar procedure has a $\Theta(k^{2/3})$ gap. Independently, unfolding relaxations can discard compatibility between HT ranks; even fully optimized all-edge allocation with scalar operator splits can have an unbounded gap. Coupling cuts improves some cases but remains an incomplete relaxation. Neither stronger operator metrics nor more center iterations alone can remove every general-HT gap. This distinction suggests allocating effort according to whether uncertainty comes from the operator minorant, omitted spectral energy, or incompatible cut maximizers.

The present implementation has substantial limits. Its general-HT tests use small dense coefficient spaces and enumerate a finite balanced-tree collection. They do not address arbitrary tree shapes or joint rank allocation. The compressed-input complexity assumes supplied spectral CP factors; discovering those factors from dense data is excluded. Variable potentials are studied in four dimensions, while the eight-mode search uses a separable operator. Ordinary floating-point guards support numerical evidence, but do not constitute interval-arithmetic verification. A certified family need not make an arbitrary downstream local optimizer converge to that family's best member. The optional excess-error certificate addresses the accuracy of the computed tensor when its own primal–dual gap is small; it does not establish convergence for every input.

The runtime evidence is also conditional. Direct certificates improve coverage and some measured selection costs, but difficult comparisons can require expensive dual optimization. The strongest shared-subtree HT heuristic is faster in the current eight-mode study. Comparisons with external compressed HT/TT solvers, end-to-end cost at matched PDE accuracy, and application-derived right-hand sides remain necessary to establish broader practical value. Repeated runs quantify timing variability on the specified inputs, not success probabilities over an application population.

The affine-forcing study supplies a particularly strong control for amortization claims. With a fixed operator and only three forcing channels, three offline solves span the complete solution family. Attaching the same transport rule to a solve-first certificate is faster than the direct reuse method in the reported sequences. Near family transitions, direct bounds certify a few queries on which the competing certificate abstains, at extra cost. A warm heuristic is cheaper still in most cases and can retain a suboptimal family for a query while incurring only a small extra energy error. Thus strict family certification and practically significant solution accuracy remain distinct objectives. Variable operators, many forcing directions, or expensive full-space solves could change this balance, but they are not established advantages of the present implementation.

The literature comparison also limits the novelty claim. Center optimization uses known rank-constrained convex envelopes, and the perturbation estimate is support-function continuity. The sharper candidate contribution is the specified compression-resistant separation, its leading constant and Dirichlet realization, together with consequences for a certifiable tree-selection procedure. Establishing priority for those precise statements requires specialist review beyond the targeted search recorded here.

10 Conclusion

We developed product-metric upper bounds for attainable energy and used them to compare prescribed HT families without requiring a full PDE solution. The balanced geometric construction provably dominates the scalar realization and remains exact on a multiscale family for which optimized scalar spectral compression with a full-energy tail retains an unbounded gap. An all-edge obstruction limits what scalar forcing allocation can recover. Gain intervals support both strict family selection and an excess-error guarantee for a computed tensor. Controlled experiments verify these comparisons and expose the computational tradeoffs. Parameter transport can amortize certification, while equally reusable solve-first baselines remain essential to assessing its value. Larger-scale application tests remain necessary to establish the practical benefit.

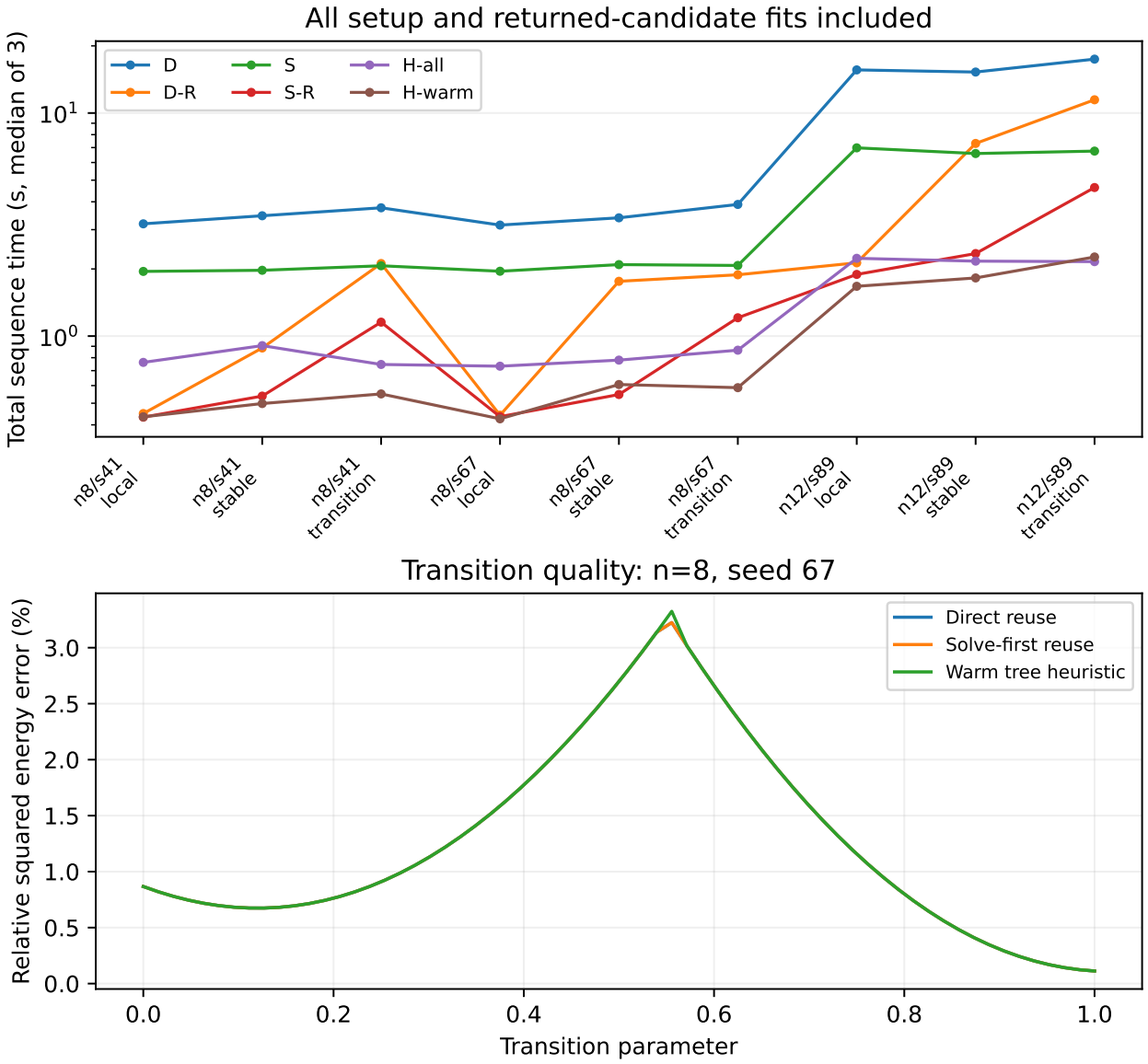


Figure 3: Complete sequence costs for the six methods, and returned-tensor quality on the $n = 8$, seed-67 transition path. The $n = 8$ sequences have 64 queries and the $n = 12$ sequences have 32. Setup and final fits are included in the timing comparison. The error comparison uses relative squared energy error; agreement in approximation quality does not imply that a heuristic supplies the strict-family guarantee.

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A Proofs of the gain bounds

All spaces below are finite-dimensional and real. Matrix inner products are Frobenius inner products after an orthonormal matricization. For a positive semidefinite operator $\mathcal{C}$, write $\|z\|_{\mathcal{C}}^2 = \langle z, \mathcal{C}z\rangle$; this can be a seminorm.

A.1 Energy identity and feasible directions

Since $\mathcal{A}u = f$, direct expansion gives

$$\|u - x\|_{\mathcal{A}}^2 = \|u\|_{\mathcal{A}}^2 - 2\langle f, x\rangle + \langle x, \mathcal{A}x\rangle.$$

Taking the infimum over $\mathcal{K}_T$ proves (4). For $x \neq 0$, the scalar quadratic $g(tx) = 2t\langle f, x\rangle - t^2\langle x, \mathcal{A}x\rangle$ is maximized at $t = \langle f, x\rangle / \langle x, \mathcal{A}x\rangle$, with value (5). Scaling preserves every rank bound. Finally, (6) implies $G_{T_*} \geq \ell_{T_*} > U_T \geq G_T$ for each competitor, proving strict optimality among the candidate families.

A.2 Proof of Proposition 4.2

For $\mathcal{C} = \mathcal{A} - \mathcal{B} \succeq 0$, completing a square yields

$$\begin{aligned} g(X) &= \langle Y, \mathcal{C}Y\rangle + 2\langle F - \mathcal{C}Y, X\rangle - \langle X, \mathcal{B}X\rangle - \|X - Y\|_{\mathcal{C}}^2 \\ &\leq \langle Y, \mathcal{C}Y\rangle + 2\langle F - \mathcal{C}Y, X\rangle - \langle X, \mathcal{B}X\rangle. \end{aligned}$$

Set $Z = L^{1/2}XR^{1/2}$ and $M = L^{-1/2}(F - \mathcal{C}Y)R^{-1/2}$. The change of variables is invertible and preserves rank. Therefore the remaining supremum is

$$\sup_{\operatorname{rank}(Z)\leq r} \{2\langle M, Z\rangle - \|Z\|_F^2\} = \|M\|_F^2 - \inf_{\operatorname{rank}(Z)\leq r} \|M - Z\|_F^2 = \mathcal{H}_r(M).$$

The last equality is the truncated-SVD characterization of the best Frobenius approximation. This proves (9); at $Y = 0$ the maximizing $Z = \mathcal{P}_r(M)$ gives (10). Convexity in Y follows because $\langle Y, \mathcal{C}Y\rangle$ is convex and $\mathcal{H}_r$ is the squared Ky Fan $(2, r)$ norm, hence convex, composed with an affine map.

A.3 Proof of Theorem 4.3

Expanding the right-hand side of (13) gives

$$\langle Y, (\mathcal{A} - \mathcal{B})Y\rangle + \sup_{\operatorname{rank}(X)\leq r} \{2\langle F - (\mathcal{A} - \mathcal{B})Y, X\rangle - \langle X, \mathcal{B}X\rangle\},$$

which equals (9). If $\mathcal{B}_2 \succeq \mathcal{B}_1$, its displacement penalty is at least as large for every $X - Y$. Taking the supremum proves (11). Taking infima over the same center set preserves the inequality.

Diagonalize the two Kronecker-sum factors. Their shifted eigenvalues $a_i + s$ and $b_j - s$ are both at least $m/2$. The corresponding product-metric eigenvalue satisfies

$$m \leq 2\sqrt{(a_i + s)(b_j - s)} \leq a_i + b_j,$$

where the second inequality follows from

$$a_i + b_j - 2\sqrt{(a_i + s)(b_j - s)} = \left(\sqrt{a_i + s} - \sqrt{b_j - s}\right)^2.$$

Thus $mI \preceq \mathcal{B}_* \preceq \mathcal{A}$. The minimum eigenvalue of the Kronecker sum is $a_0 + b_0 = m$, so mI is its largest scalar minorant.

A.4 Proof of Theorem 4.4

On the high-mode block of (14), both $\mathcal{A}$ and $\mathcal{B}_*$ act by multiplication by D. The forcing vanishes elsewhere. The weighted forcing has k singular values $D^{-1/2}$, so the product bound at zero is r/D. A diagonal matrix with value $1/D$ on any r high coordinates attains that gain. The feasible lower bound and the product upper bound coincide, proving (15).

For the scalar metric, put $\mathcal{C} = \mathcal{A} - mI$ and let P project onto the k high coordinates. Replacing Y by PYP cannot increase its $\mathcal{C}$-quadratic term, because $\mathcal{C}$ is diagonal with nonnegative coefficients in the matrix-unit basis. Also

$$F - \mathcal{C}(PYP) = P(F - \mathcal{C}Y)P.$$

Orthogonal row and column projections cannot increase singular values, so they cannot increase $\mathcal{H}_r$. It therefore suffices to consider centers on the high block, where $\mathcal{C}$ acts by $d = D - m > 0$ and $F = I_k$.

The resulting convex objective $d\,\|Y\|_F^2 + m^{-1}\mathcal{H}_r(I_k - dY)$ is invariant under simultaneous orthogonal conjugation. Averaging Y over signed permutation matrices eliminates its off-diagonal entries and equalizes the diagonal, without increasing the objective. Its infimum is thus attained at a center tI_k. Minimizing

$$dkt^2 + \frac{r}{m}(1 - dt)^2$$

gives $t = r/(mk + rd)$ and value $kr/(mk + rd)$, which is (16). Dividing by r/D yields the stated ratio. For $D/m = k^2$ and fixed r, the ratio grows asymptotically as k/r.

A.5 Proof of Corollary 4.5

We first establish a nonuniform-band estimate. Suppose the two cut factors have identical spectra (a_i) and the spectral forcing is diagonal with unit entries on a set S of size $k \geq r$. Put $d_i = 2a_i$ on S, $\bar{d} = k^{-1}\sum_{i\in S} d_i$, and $m = 2\min_i a_i$. The balanced metric is exact on each diagonal matrix unit. Truncating the r largest weighted forcing singular values therefore proves

$$G_r = \Phi_{\mathcal{B}_*,r}(0) = \sum_{i\in I_r} \frac{1}{d_i}, \tag{51}$$

where I_r contains the r smallest d_i in S.

Let P_S denote the diagonal projector onto S, and sample a uniformly random r-element subset $J \subset S$. For any real t, the matrices $X_J = (k/r)t\,\mathrm{diag}(\mathbf{1}_J)$ are rank-r feasible and satisfy

$$\mathbb{E}X_J = tP_S, \qquad \mathbb{E}\,\|X_J\|_F^2 = (k^2/r)t^2.$$

A supremum is at least the average over these matrices. With $\mathcal{C} = \mathcal{A} - mI$, this gives, for every center,

$$\begin{aligned}\Phi_{mI,r}(Y) &\geq \langle Y, \mathcal{C}Y\rangle + 2\langle F - \mathcal{C}Y, tP_S\rangle - m(k^2/r)t^2 \\ &= \|Y - tP_S\|_{\mathcal{C}}^2 + 2kt - [k\bar{d} + mk(k/r - 1)]t^2.\end{aligned}$$

Discarding the first term and maximizing over t proves

$$\inf_Y \Phi_{mI,r}(Y) \geq \frac{k}{\bar{d} + m(k/r - 1)}. \tag{52}$$

If $D \leq d_i \leq \rho D$ throughout the band, then (51) is at most r/D, hence the ratio is at least $(k/r)D/[\rho D + m(k/r - 1)]$.

For the specified Dirichlet problem, the two cut eigenvalues are $a_{(p,q)} = \lambda_p + \lambda_q$. On $p, q > n/2$, there are $k = n^2/4$ selected pairs, and

$$D := 8(n+1)^2 < d_{(p,q)} < 16(n+1)^2 = 2D, \qquad m = 4\lambda_1 \leq 4\pi^2.$$

The preceding estimate therefore gives

$$\frac{\inf_Y \Phi_{mI,r}(Y)}{G_r} \geq \frac{2n^2(n+1)^2}{16r(n+1)^2 + \pi^2 n^2} \geq \frac{2n^2}{16r + \pi^2},$$

as required. Scaling the forcing changes both gains quadratically and leaves the ratio unchanged.

A.6 Proof of Theorem 5.1

A product projector acts on either side of every matricization by left and right multiplication, so it cannot increase a matricization rank. Thus $x \in \mathcal{K}_T$ implies $Px \in \mathcal{K}_T$. Since P is orthogonal and commutes with $\mathcal{A}$, writing $x_P = Px$ and $x_Q = Qx$ gives the exact splitting

$$g(x) = 2\langle Pf, x_P\rangle - \langle x_P, \mathcal{A}x_P\rangle + 2\langle h, x_Q\rangle - \langle x_Q, \mathcal{A}x_Q\rangle.$$

The first term is at most $G_{T,P}$. Allowing arbitrary x_Q can only increase the second term, whose supremum is $\langle h, \mathcal{A}^{-1}h\rangle$ by completing the square. Taking the supremum proves the upper enclosure. Restricting to the retained feasible family proves the lower one. If $f = Pf$, the omitted term is nonpositive and setting $x_Q = 0$ is feasible, proving the full-space equality for retained forcing. The certificate (20) follows by inserting the resulting intervals into (6).

The coefficient one on E_{tail} is sharp. In two modes, retain the first basis vector in each mode, use diagonal axis operators, and choose $f = a\, e_1 \otimes e_1 + b\, e_2 \otimes e_1$ with $b \neq 0$. The entire solution has matrix rank one, as does its retained part. With rank budget one, the full gain therefore equals retained gain plus omitted energy.

A.7 Proof of Proposition 5.5

Use three modes of dimension n^2, label each coordinate by an ordered pair, and define the matrix-multiplication tensor by

$$F_{(i,j),(j,k),(k,i)} = 1 \quad (1 \le i, j, k \le n),$$

with all other entries zero. With all tree ranks equal to one, the feasible family consists of fully separable tensors. Reshape three unit mode vectors as matrices X, Y, Z of unit Frobenius norm. Then

$$\langle F, X \otimes Y \otimes Z\rangle = \operatorname{tr}(XYZ), \qquad |\operatorname{tr}(XYZ)| \le \|X\|_F \|Y\|_F \|Z\|_F = 1.$$

Compatible rank-one matrix units attain equality. Optimizing the scalar multiplier therefore gives $G_T = 1$ for $\mathcal{A} = I$.

Every singleton unfolding has mutually orthogonal rows, each containing exactly n unit entries. All its nonzero squared singular values equal n, so its rank-one quadratic gain is n. These singleton constraints are all the distinct edges of a three-mode binary tree, up to complements. For the exact product metric $\mathcal{B} = \mathcal{A} = I$, the center correction is zero; changing centers cannot improve any edge value. This proves the unbounded gap.

A.8 Proof of Proposition 5.6 and the example

The allocations (31) satisfy $f_1 + f_2 = f$ and $\mathcal{A}_1 + \mathcal{A}_2 = \mathcal{A}$. Consequently the objective splits exactly. Every HT-feasible x belongs to both cut families, so each split objective is at most its own centered upper bound. Adding and taking the supremum proves (32).

Each term is a supremum of affine functions of its forcing argument plus a constant, proving convexity in D. Differentiating these suprema where the maximizers are unique gives $2x_1 - 2x_2$. More generally, the affine function associated with any maximizing pair supports the supremum, giving the same expression as a subgradient. Restricting the admissible allocations to a coordinate subspace preserves validity of every bound, though it may worsen the minimum.

For the binary example, each singleton unfolding of f has squared singular values $2, 1$. Under the stated allocation, the first allocated forcing has two orthogonal rows with squared norms

$$(5/8)^2 = (3/8)^2 + (1/2)^2 = 25/64.$$

The same is true in the second cut for its allocated forcing. Taking the exact metrics $\mathcal{B}_1 = \mathcal{B}_2 = I/2$ yields two contributions $2(25/64) = 25/32$, hence $J(D) = 25/16$.

This allocation is globally optimal even when the split varies. Let $\alpha_1 = \alpha$, $\alpha_2 = 1 - \alpha$, and denote the nuclear norm by $\|\cdot\|_*$. Spectral/nuclear norm duality and the scalar square inequality imply, for every tensor Z,

$$\frac{\left\|f_i^{(i)}\right\|_{\mathrm{op}}^2}{\alpha_i} \geq 2\langle f_i, Z\rangle - \alpha_i \left\|Z^{(i)}\right\|_*^2, \qquad i = 1, 2.$$

Choose $Z_{100} = Z_{010} = 15/32$, $Z_{001} = 20/32$, and all other entries zero. In either cut, its two rows are orthogonal with lengths $15/32$ and $25/32$, giving $\left\|Z^{(i)}\right\|_* = 5/4$. Also $\langle f, Z\rangle = 25/16$. Adding the preceding inequalities therefore proves

$$J_\alpha(D) \geq 2\langle f, Z\rangle - \sum_{i=1}^{2} \alpha_i \left\|Z^{(i)}\right\|_*^2 = \frac{25}{16}$$

for every full-tensor allocation D and every $0 < \alpha < 1$. The explicit allocation attains this lower bound, proving the optimized value in (33).

Finally, rank one in the first two singleton cuts already forces complete separability. For a fixed unit third vector (e, t), maximizing the contraction over the first two unit vectors gives the spectral norm of

$$\begin{pmatrix} t & e \\ e & 0 \end{pmatrix}.$$

With $e^2 + t^2 = 1$, this norm equals $(|t| + \sqrt{4 - 3t^2})/2$. Its maximum on $|t| \leq 1$ is $2/\sqrt{3}$, attained at $|t| = 1/\sqrt{3}$. Squaring proves the true gain $4/3$ in (33).

A.9 Proof of Proposition 5.2

Put $\mathcal{C} = \mathcal{A} - mI \succeq 0$. For every diagonal sign matrix $S = \mathrm{diag}(\varepsilon_1, \ldots, \varepsilon_n)$, with $\varepsilon_i \in \{-1, 1\}$, the scalar centered objective is invariant under $Y \mapsto SYS$. Indeed, $SFS = F$, $\mathcal{C}(SYS) = S(\mathcal{C}Y)S$, and orthogonal conjugation preserves the quadratic term and singular values. Averaging over these 2^n sign matrices gives $\mathrm{diag}(Y_{11}, \ldots, Y_{nn})$. Convexity of $\Phi_{mI,r}$ therefore shows that its infimum can be taken over diagonal centers $Y = \mathrm{diag}(y_1, \ldots, y_n)$.

Write $c_i = d_i - m \geq 0$. The sum of the largest r members of a nonnegative list $z_1, \ldots, z_n$ is $\max_{p \in \Delta_{n,r}} \sum_i p_i z_i$. Consequently the remaining optimization is

$$\inf_{y \in \mathbb{R}^n} \max_{p \in \Delta_{n,r}} Q(y, p), \qquad Q(y, p) = \sum_{i=1}^{n} \left[c_i y_i^2 + \frac{p_i}{m}(f_i - c_i y_i)^2\right]. \tag{53}$$

There is no invertibility assumption on $\mathcal{C}$ in this reduction. If $c_i = 0$, the ith summand is independent of y_i, so fix $y_i = 0$. If $c_i > 0$, its minimizer for a fixed $p_i \in [0, 1]$ is

$$y_i(p_i) = \frac{p_i f_i}{m + c_i p_i}.$$

Since $0 \leq p_i/(m + c_i p_i) \leq 1/d_i$, all such minimizers lie in the closed interval with endpoints 0 and f_i/d_i. Projecting any y_i onto that interval cannot increase the summand for any p_i. Thus the infimum in (53) may be restricted to the compact convex product K of these intervals and the singleton coordinates $\{0\}$ where $c_i = 0$.

The function Q is continuous on $K \times \Delta_{n,r}$, convex in y, and affine in p. Both domains are nonempty compact convex sets, so the compact convex–concave minimax theorem gives

$$\min_{y \in K} \max_{p \in \Delta_{n,r}} Q(y, p) = \max_{p \in \Delta_{n,r}} \min_{y \in K} Q(y, p).$$

For $c_i > 0$, substitution of $y_i(p_i)$ gives the minimum $f_i^2 p_i/(m + c_i p_i)$. For $c_i = 0$, the same expression is $f_i^2 p_i/m$ and is already the constant summand. Summing proves (21). Finally, each denominator is $d_i p_i + m(1 - p_i)$, which is nondecreasing in m; its numerator is nonnegative. Taking the maximum preserves the claimed monotonicity.

A.10 Proof of Lemma 5.3

Put $J = L \cap R$. The retained forcing is diagonal with nonzero entries exactly on J, and every omitted diagonal entry has energy $f_i^2/d_i = 1$. Hence $E(L,R) = k - |J|$. If $J = \varnothing$, the retained forcing is zero and its optimized bound is zero, so the certificate is k, also obtained from $J_k = \varnothing$.

Assume $J \neq \varnothing$ and put $m = m_{L,R}$. For centers in the $L \times R$ rectangle, replacing Y by $P_J Y P_J$ cannot increase the scalar centered bound. The quadratic term decreases because $\mathcal{A} - mI$ is diagonal with nonnegative coefficients in the matrix-unit basis. For the residual $M = \mathcal{P}_{L,R}F - (\mathcal{A} - mI)Y$, the new residual is $P_J M P_J$; orthogonal row and column projections cannot increase its largest singular value. The restricted scalar infimum is therefore exactly the infimum on the $J \times J$ block with the same scalar m.

The best scalar on the matched block is $m_J = \min_{i \in J} d_i$. Since $m_{L,R} \leq m_J$, Proposition 5.2 shows that replacing L, R by J, J cannot increase the optimized scalar bound. Its tail is unchanged. This proves the first reduction.

Now fix a nonempty matched set J and let $j = \min J$, so its scalar is $m = d_j$. If $i > j$ is absent, add it without changing m. In the maximization (21), the new term is bounded by one, because

$$0 \leq \frac{d_i p_i}{m + (d_i - m)p_i} \leq 1 \qquad (0 \leq p_i \leq 1).$$

For every feasible allocation on $J \cup \{i\}$, its restriction to J is feasible on J. Thus the optimized scalar bound increases by at most one upon adding i, whereas the tail decreases by exactly one. Filling all missing indices above j consequently cannot increase the total certificate. The resulting set is J_{j-1}. The formula for its scalar bound is (21) with $r = 1$, $m = d_{s+1}$, and $f_i^2 = d_i$; its tail is s. This proves both remaining claims.

A.11 Proof of Theorem 5.4

Write $T_k = \mathrm{diag}(1, \ldots, k)$. The balanced metric is $\mathcal{B}_*(X) = T_k X T_k$. Its whitened forcing satisfies

$$T_k^{-1/2} F_k T_k^{-1/2} = I_k.$$

Hence $\Phi_{\mathcal{B}_*,1}(0) = 1$. For every i, the rank-one matrix $X_i = e_i e_i^{\mathsf{T}}/i$ has $\langle F_k, X_i \rangle = 1$ and $\langle X_i, \mathcal{A} X_i \rangle = 1$, giving gain one. The feasible gain and the product upper bound coincide, proving (28).

Every diagonal forcing mode has full-solution energy one. By Lemma 5.3, $V_k = \min_s (s + U_s)$, with $U_k = 0$, and for $s < k$ the retained scalar is $m_s = (s+1)^2$. For the upper bound put $a = k^{2/3}$ and $s = \lceil a \rceil < k$. The zero center on that suffix yields

$$U_s \leq \Phi_{m_s I,1}(0) = \frac{k^2}{(s+1)^2} \leq a.$$

Therefore $V_k \leq s + U_s \leq 2a + 1 \leq 3a$.

For the lower bound, take any $s \in \{0, \ldots, k\}$. If $s \geq a/4$, nonnegativity gives $s + U_s \geq a/4$. Otherwise $s < a/4 < k/2$, and every index in $H = \{k/2 + 1, \ldots, k\}$ remains in the suffix. Let $n = k/2$ and use the feasible allocation $p_i = 1/n$ on H and zero elsewhere in (26). With $b = k^2/4$ we have $d_i = i^2 \geq b$ on H, and hence

$$U_s \geq \sum_{i \in H} \frac{d_i}{n m_s + d_i - m_s} \geq \sum_{i \in H} \frac{d_i}{n m_s + d_i} \geq \frac{nb}{n m_s + b} = \frac{k^2}{4m_s + 2k}.$$

Since $k \geq 64$, we have $a \geq 16$, $k \leq a^2/4$, and $s + 1 \leq a/4 + 1 \leq a/2$. Therefore $m_s \leq a^2/4$ and $4m_s + 2k \leq 3a^2/2$. It follows that $U_s \geq 2k^2/(3a^2) = 2a/3 \geq a/4$. Both cases give $s + U_s \geq a/4$, and taking the minimum proves the lower bound in (29).

For completeness, no other commuting orthogonal product projectors are missing from the comparison. A nonzero product range has the form $\mathcal{L} \otimes \mathcal{R}$. If it is invariant under $A_L \otimes I + I \otimes A_R$, then applying the projection onto $\mathcal{L}^\perp \otimes \mathcal{R}$ to the image of $x \otimes y$, for $x \in \mathcal{L}$ and nonzero $y \in \mathcal{R}$, shows $A_L \mathcal{L} \subseteq \mathcal{L}$. Likewise $A_R \mathcal{R} \subseteq \mathcal{R}$. The two symmetric factors have simple spectra, so their invariant subspaces are spans of subsets of the displayed coordinate eigenvectors. Thus every such projector is one of the $\mathcal{P}_{L,R}$ already considered; the zero projector is also included. Finally, the eigenvalues of $\mathcal{A}$ are $(i^2 + j^2)/2$, with minimum one and maximum k^2.

B Proofs for parameter reuse

B.1 Proof of Proposition 7.1

For nonzero $x \in \mathcal{K}_T$, optimizing its real amplitude gives

$$\max_{a\in\mathbb{R}}\{2a\langle f, x\rangle - a^2 \|x\|_{\mathcal{A}}^2\} = \frac{|\langle f, x\rangle|^2}{\|x\|_{\mathcal{A}}^2}.$$

Taking the supremum over feasible directions and using the cone property proves the support-function representation. If the cone is $\{0\}$, both sides are zero. The feasible unit ball is compact in finite dimensions; its symmetry under $x \mapsto -x$ also allows removal of the absolute value when interpreting this expression as a support function. For every $\|x\|_{\mathcal{A}} \le 1$,

$$\big||\langle f, x\rangle| - |\langle h, x\rangle|\big| \le |\langle f - h, x\rangle| \le \|f - h\|_{\mathcal{A}^{-1}}.$$

Taking suprema proves the one-sided inequality, and exchanging f, h proves the other side. Since $s_T(tf_0) = |t| s_T(f_0)$, the homogeneous transport inequality follows by choosing $h = tf_0$. Inserting the anchor bounds and using nonnegativity proves (40).

The operator inequality $\mathcal{B} \preceq \mathcal{A}$ implies $\mathcal{A}^{-1} \preceq \mathcal{B}^{-1}$. Applying it to $f(\theta) - tf(\theta_a)$ and expanding the inverse-$\mathcal{B}$ quadratic form gives (42). This also proves that Q is positive semidefinite. Minimization of this quadratic polynomial in t gives (43) when its leading coefficient is positive. Otherwise $\|f(\theta_a)\|_{\mathcal{B}^{-1}} = 0$, so the anchor forcing is zero and all choices of t have the same residual. Dependent forcing basis vectors therefore cause no mathematical singularity.

B.2 Proof of Corollary 7.2

Each transported interval contains the same $s_T(f(\theta))$, so their intersection does too. Optimizing the amplitude of a retained feasible direction gives (44) and (45); these provide additional valid lower endpoints. Strict separation of the root gains implies strict separation of their squares, because every endpoint used in the test is nonnegative. Equation (4) then proves unique family optimality. Applying Proposition 4.1 with $U_T = \overline{s}_T^2$ and the actual gain of the returned tensor gives (47). None of these arguments assumes that an anchor or feasible candidate was selected independently of earlier queries.

C Sharper consequences of the compression example

We record two refinements of Theorem 5.4. They concern the same comparator: one commuting product projection, an optimized scalar centered bound on its range, and the exact unrestricted full-energy tail. They make no statement about tensor-tree selection time.

Proposition C.1 (Sharp asymptotic constant)**.** *For V_k in (24), with $d_i = i^2$ and $f_i = i$,*

$$\lim_{k\to\infty} \frac{V_k}{k^{2/3}} = \frac{3}{2^{2/3}}. \tag{54}$$

Proof. By Lemma 5.3, it suffices to keep a suffix $\{a, \dots, k\}$. Sign averaging, as in Appendix A.9, reduces the centered scalar objective to diagonal centers. Introduce a threshold z for the largest absolute residual coordinate. The coordinate a has zero center penalty coefficient and forces $z \ge a$. For $i > a$, the smallest center penalty with residual at most z is $(i - z)_+^2/(i^2 - a^2)$. Thus

$$V_k = \min_{1\le a\le k} \min_{a\le z\le k} \left\{ a - 1 + \frac{z^2}{a^2} + \sum_{i=a+1}^{k} \frac{(i - z)_+^2}{i^2 - a^2} \right\}. \tag{55}$$

Values $z > k$ cannot improve the objective. The empty suffix has value k, already attained by $a = z = k$.

Take $z = k$ and choose an integer a nearest to $2^{1/3}k^{2/3}$. For all sufficiently large k this is an admissible choice, and it proves $\limsup V_k/k^{2/3} \le 3/2^{2/3}$. Let (a_k, z_k) be minimizing pairs in (55). The preceding upper bound and

$$\sum_{i=a_k+1}^{k} \frac{(i-z_k)_+^2}{i^2-a_k^2} \ge \frac{1}{k^2}\sum_{i>z_k}(i-z_k)^2 \ge \frac{(k-z_k)^3}{3k^2}$$

imply $k - z_k = O(k^{8/9}) = o(k)$. The last inequality compares the sum of an increasing function with its integral on $[z_k, k]$. Dropping the nonnegative sum and minimizing over all positive real a now gives

$$V_k \ge a_k - 1 + \frac{z_k^2}{a_k^2} \ge \frac{3}{2^{2/3}} z_k^{2/3} - 1.$$

Since $z_k/k \to 1$, this proves the matching lower limit. □

Corollary C.2 (Ordinary two-dimensional Dirichlet discretization)**.** *Let T_k be the standard second-order Dirichlet difference matrix with k interior points on $(0,1)$, and let $\mathcal{A}(X) = T_kX + XT_k$. In the orthonormal sine basis its one-dimensional eigenvalues are*

$$\lambda_i = 4(k+1)^2 \sin^2\left(\frac{i\pi}{2(k+1)}\right).$$

Specify the forcing in that basis as $F = \operatorname{diag}(\sqrt{2\lambda_1}, \dots, \sqrt{2\lambda_k})$, and use rank one. Then $G_1 = \Phi_{\mathcal{B}_*,1}(0) = 1$. The optimized scalar-plus-exact-tail certificate $V_{k,D}$ satisfies, for every integer $k \ge 1$,

$$\frac{1}{4\pi^2}k^{2/3} \le V_{k,D} \le \left(1 + \frac{\pi^2}{4}\right)k^{2/3}. \tag{56}$$

Proof. More generally, let the strictly increasing d_i satisfy $ci^2 \le d_i \le Ci^2$ for fixed $0 < c \le C$, and take $F = \operatorname{diag}(\sqrt{d_i})$ and $\mathcal{A}(X) = (DX + XD)/2$. The balanced product metric has coefficients $\sqrt{d_i d_j}$, so its whitened forcing is the identity. The feasible matrix $e_i e_i^{\mathsf{T}}/\sqrt{d_i}$ has gain one, establishing exactness of the product bound. Each omitted diagonal forcing mode has energy one. The suffix reduction remains valid, and the threshold argument above gives

$$V_k(d) = \min_{1\le a\le k} \min_{\sqrt{d_a}\le z\le\sqrt{d_k}} \left\{ a - 1 + \frac{z^2}{d_a} + \sum_{i=a+1}^{k} \frac{(\sqrt{d_i}-z)_+^2}{d_i - d_a} \right\}.$$

Set $a = \lceil k^{2/3} \rceil$ and $z = \sqrt{d_k}$. This gives the upper bound $(1 + C/c)k^{2/3}$. For the lower bound, consider any admissible (a, z). If $z < \sqrt{c}\,k/4$, then $a < k/4$. Every index $i \ge \lceil k/2 \rceil$ is therefore present and contributes at least $1/4$ to the sum, since $\sqrt{d_i} \ge \sqrt{c}\,k/2$. There are at least $k/2$ such indices, giving a lower bound $k/8$.

Otherwise $z \ge \sqrt{c}\,k/4$. If $a = 1$, the threshold term is at least $(c/C)k^2/16$. If $a \ge 2$, put $r = c/C \le 1$ and use

$$a - 1 + \frac{z^2}{d_a} \ge \frac{a}{2} + \frac{rk^2}{16a^2} \ge \frac{3r^{1/3}}{4\,2^{2/3}}k^{2/3} \ge \frac{r}{16}k^{2/3}.$$

All cases give $\frac{c}{16C}k^{2/3} \le V_k(d) \le (1 + C/c)k^{2/3}$. For the stated difference operator, the elementary bounds $(2/\pi)x \le \sin x \le x$ on $[0, \pi/2]$ imply $4i^2 \le \lambda_i \le \pi^2 i^2$. Taking $d_i = 2\lambda_i$, $c = 8$, and $C = 2\pi^2$ gives (56). The simple factor spectra also ensure that every commuting orthogonal product projector is covered by the coordinate-subset comparison, as in Appendix A.11. □

The forcing here depends on the discretization and is manufactured to expose the relaxation gap. The corollary is not a statement about a fixed smooth continuum load or typical application forcing.

k	V_k	$V_k/k^{2/3}$	$V_{k,D}$	$V_{k,D}/k^{2/3}$
32	13.11954	1.30162	11.36597	1.12765
128	36.64430	1.44278	30.83649	1.21411
256	60.38199	1.49766	50.19660	1.24503
1024	161.04317	1.58517	131.09047	1.29034
2048	261.30377	1.62029	210.80639	1.30717

Table 6: Optimized scalar certificates with exact full-energy tails. The product bound and actual rank-one gain equal one in both constructions. The polynomial case has proved limiting normalized value $3/2^{2/3} \approx 1.88988$; convergence is slow at these sizes. The Dirichlet corollary proves the order, not that same limiting constant. Values are obtained by the scalar convex threshold minimization above. An independent exhaustive subset check for $k = 2, \ldots, 10$ agrees with the suffix reduction. These are evaluations of bounds, not PDE runtime measurements.

D Experimental protocols and reproducibility

D.1 Scope, accounting, and environment

All norms and inner products refer to the finite-dimensional discretized problem. We do not estimate continuum discretization errors. The recorded experiments use Python 3.11.2, NumPy 2.4.6, and SciPy 1.17.1. The HT run records Linux on an aarch64 host with glibc 2.36; a CPU model was not recorded, so we do not infer one. The accompanying environment pins Matplotlib 3.11.2 for figure generation. Both `OPENBLAS_NUM_THREADS` and `OMP_NUM_THREADS` are set to one.

Four-dimensional and HT comparisons repeat complete method calls three times, with deterministically shuffled method order using seed 20260920. The reported per-case time is the median of these calls. CP timings use seven complete calls following one warm-up. Input construction is excluded from all timings; transformations, operator setup, cache construction, bound evaluations, local fits, fallback stages, and decision checks are included. HT tree enumeration is shared and warmed before timing. Independent post-timing baseline edge-rank diagnostics are excluded from the HT timings. No confidence interval or population success probability is inferred from these few repeats or from the deliberately structured problem sets.

Raw results retain failed certificates, bounds, selected trees, work counts, and timings for each repeat. Source-file SHA-256 hashes are stored in the four-dimensional and HT configurations. The script `paper/generate_tables.py` derives the manuscript tables directly from the recorded JSON data, checks decision consistency and certificate agreement, and writes a machine-readable fact sheet with hashes of its inputs. It does not rerun benchmarks or replace the recorded measurements.

D.2 Four-dimensional source and operator construction

Let $x_i = k_i/(n+1)$ for $k_i \in \{1, \ldots, n\}$ and let L_i be the standard second-order Dirichlet difference operator acting on coordinate i. For a seeded permutation p of the four axes and a seeded $\beta \sim \mathrm{Unif}[7, 13]$, define

$$\widetilde{f}_{\mathrm{low}}(x) = \prod_{i=1}^{4} \sin(\pi x_i) \exp\left[-\beta\left((x_{p_1} - x_{p_2})^2 + (x_{p_3} - x_{p_4})^2\right)\right],$$

$$\widetilde{f}_{\mathrm{high}}(x) = \prod_{i=1}^{4} \sin(3\pi x_i) \exp\left[-\beta\left((x_{p_1} - x_{p_3})^2 + (x_{p_2} - x_{p_4})^2\right)\right].$$

Each component is normalized in the unweighted Euclidean grid norm. The paired family uses $f = f_{\mathrm{low}}$; the competing family uses $f = f_{\mathrm{low}} + 4f_{\mathrm{high}}$. Both have $a_i = 1$ and $V = 0$. The anisotropic family uses the same mixture and diffusion coefficients $(a_{p_1}, a_{p_2}, a_{p_3}, a_{p_4}) = (1, 2, 5, 9)$. The potential family uses unit diffusion and

$$V(x) = 10 + 120\left[1 + \tfrac{1}{2}\sin\big(2\pi(x_{p_1} + x_{p_3})\big) + \tfrac{1}{2}\cos\big(2\pi(x_{p_2} - x_{p_4})\big)\right]^2.$$

For the development group, paired sources use seeds $0, 1, 2$, and each other family uses seeds $0, 1$, giving nine PDE instances and 18 comparisons at $r = 1, 2$. The fresh-seed group substitutes seeds $11, 17, 29$, with only

the first two used outside the paired family. It has the same size and uses the same generator. Grid-scaling groups use the paired, competing, and potential families with seed zero, $r = 1$, and $n = 12, 16$.

The sensitivity study fixes seed zero and uses $f = f_{\text{low}} + \alpha f_{\text{high}}$ for $\alpha \in \{0, 0.25, 0.5, 1, 1.5, 2, 3, 4, 6, 8\}$, with either zero potential or the potential above, at $n = 8$ and $r = 1$. Its control is $\prod_i \sin(\pi x_i) + 0.3 \prod_i \sin(2\pi x_i)$ with zero potential; symmetry makes all three candidate gains equal. Development and sensitivity cases informed implementation choices. Some cases overlap across groups, and ranks one and two reuse PDE instances. The 63 reported case/rank rows are therefore not 63 independent random PDEs.

Adaptive policy. The initial pass retains the four lowest one-dimensional modes and uses zero local-ALS sweeps. For constant potential, an unresolved case next retains five modes and uses successive per-stage sweep caps $0, 2, 6$. Remaining cases use full-grid balanced-geometric bounds with budgets $0, 2, 6$, followed by a richer metric pool with budgets $0, 2, 6, 16$ and up to 40 L-BFGS-B iterations for dual centers. At each stage, lower bounds are maximized and upper bounds minimized across all earlier stages; a successful separation stops the procedure. The richer pool contains the balanced-geometric metric, five scaled power products with exponents $0, 1/4, 1/2, 3/4, 1$, and three row-shift constructions. For spatially varying potential, separable operator minorants support the upper bounds, while all feasible gains are evaluated using the actual potential. This may leave a relaxation gap.

Strong solve-based comparisons. Constant-potential problems use a discrete sine transform spectral inverse. Variable-potential problems use CG preconditioned by the separable operator with the mean potential, with relative residual tolerance 10^{-10}. The proxy comparator tests the residual-distance certificate at iterations $1, 2, 4, 8, 12, 20, 32, 48, 64, 96, 120$ and at termination, stopping online if certified; its maximum is 120 iterations. Its known inconclusive zero iterate is skipped. The residual radius is $\|f - Aw\|_2/\sqrt{m}$ for the implemented coercivity lower estimate m. More explicitly, set $d_T(w) = \inf_{x \in \mathcal{K}_T} \|w - x\|_{\mathcal{A}}$ and let $\underline{d}_T(w) \le d_T(w) \le \overline{d}_T(w)$ be computable distance bounds. Distance to any nonempty set is one-Lipschitz, while $\|u - w\|_{\mathcal{A}} \le \delta := \|f - \mathcal{A}w\|_2/\sqrt{m}$. The residual-distance comparator therefore certifies S when

$$\min_{T \ne S} \underline{d}_T(w) - \overline{d}_S(w) > 2\delta.$$

The solve-then-certify variant applies this same criterion to its completed numerical solution; the online proxy variant checks intermediate iterates. Operator and potential-minorant factors are cached within each invocation. Gram eigendecompositions compute spectral-tail bounds, with a downward roundoff allowance, while leading-SVD candidates use ARPACK with tolerance 10^{-10} and fixed seed 2026. The Frobenius comparator uses the same fast complete solver and leading singular values, but no energy certificate.

D.3 CP input protocol

The CP input has eleven terms: four for a normalized paired low-band source, four for a differently paired high-band source of amplitude four, and three broad perturbation terms. Each perturbation factor is an independently generated Gaussian column normalized in Euclidean norm, using seed 20260920; the three perturbation weights are $\alpha/\sqrt{3}$. We vary $n \in \{16, 64, 256, 1024, 4096\}$ and $\alpha \in \{0, 0.1, 0.5, 1\}$. Each axis retains zero-based indices $\{0, 1, 10, 11\}$. Tail norms are accumulated through disjoint first-omitted-axis blocks and Hadamard products of small Gram matrices, avoiding subtraction of nearly equal full and retained squared norms.

At $n = 16$ only, a dense spectral solution and the earlier certificate serve as evaluation oracles after compressed selection. Their timings are not used as a headline scaling comparator. Producing or fitting the input CP factors is outside this experiment's cost model. The five tensor sizes with repeated perturbation amplitudes form a structured scaling study; the resulting 15/20 certification count is not an estimated population success rate.

D.4 All-edge HT protocol and scheduling ablation

The eight-dimensional search enumerates all 315 balanced binary trees on the labeled axes; arbitrary unbalanced topologies are outside this family. Every nonroot rank is capped at two. The small cases use

Modes per axis	Certificates	Time range (ms)	Factor coefficients
16	3/4	1.96–4.62	715
64	3/4	2.00–2.64	2,827
256	3/4	2.05–2.79	11,275
1024	3/4	2.48–3.23	45,067
4096	3/4	4.46–5.09	180,235

Table 7: Supplied-CP input accounting. Time ranges are the minimum and maximum of the four amplitude-specific median times at each n. Factor coefficients count forcing factors and weights; an additional $4n$ eigenvalues are supplied. Every run assembles only 256 retained tensor entries. The largest perturbation amplitude is unresolved at every size.

$n \in \{2, 3\}$ and seeds $31, 47, 59$. Source leaf and transfer frames are orthonormalized Gaussian matrices, with an equal-singular-value root core, and the resulting forcing is normalized. Axis order is scrambled using seed $17000 + s$, where s is the source seed.

Band cases use a 4^8 spectral tensor with one-dimensional eigenvalues $4 \cdot 65^2 \sin^2(\pi k/130)$ for $k \in \{1, 2, 11, 12\}$. A normalized rank-two HT source occupies the all-low block, and four times a differently structured source occupies the all-high block. All other entries are zero. The high tree applies order $(1, 3, 5, 7, 2, 4, 6, 8)$ to the low tree's leaf order. Seeds $101, 103, 107$ are initial development cases. Seeds $131, 137$ were first used with the late schedule and subsequently observed while developing the early policy. Seeds $151, 157, 163$ were fixed before testing that policy. All noninitial band cases use the same independently seeded axis-scrambling rule. No solution is manufactured in any of these source constructions.

The initial proxy applies the inverse of a modewise product minorant, not the inverse of the PDE operator. If needed, the preset cumulative preconditioned Richardson budgets are $0, 2, 5, 10$, with at most eight HT fits per stage. A feasible fit comprises nested HT rounding and two energy-ALS sweeps, including leaf-basis updates. The early policy attempts joint-cut refinement when one through six competitors remain after a fit. For each competitor, it combines the root cut with up to two other restrictive cuts, trying allocation fractions $0.5, 0.8, 0.95$ and at most 40 L-BFGS-B iterations per trial. These policy choices were informed by development cases. Allocation variables are restricted to the union of the nonzero source and center supports; every resulting objective value remains an upper bound. Exactly zero rows and columns may be removed before SVDs, preserving their nonzero singular values without a truncation tolerance.

The final early-schedule runs certify all non-tie cases before any Richardson step. They use one feasible HT fit each, but this does not mean one SVD or one bound evaluation. The shared-subtree baseline also applies energy ALS only once, after constructing 315 rounded candidates. Its equivalence to naive repeated rounding is checked for all candidate gains; the naive timings are not used to claim an advantage. The exact spectral solution is inexpensive in this separable model, and is formed only by the baseline.

For band seed 103, moving the same joint-bound refinement from the end of search to the early shortlist stage reduces the number of HT fits from 12 to 1 and proxy steps from 10 to 0. For seed 137, the corresponding counts change from 9 to 1 and from 10 to 0. Both late and early schedules certify these cases. These are deterministic work-count comparisons from recorded runs. Their historical timing measurements were collected in different runs and are not a paired, scheduling-only timing experiment.

D.5 Numerical checks and reproduction commands

The implementation uses a relative decision guard $10^{-9} \max\{10^{-12}, \max_T U_T\}$ for tree separation. HT feasibility checks use a relative singular-value-tail tolerance of 10^{-9}. Floating-point safeguards do not replace interval arithmetic, especially near exact ties. The excess-error rule uses $\max\{0, U - L\} + 10^{-9} \max\{10^{-12}, U, L\}$ and keeps its outcome separate from strict family separation. Optimizer convergence is not required for validity of an upper bound: the best valid objective value visited is retained even if optimization stops early. Symmetric controls must abstain from claiming a unique family, but may satisfy the separate excess-error tolerance.

The original recorded suite contains 117 passing tests, including independent energy oracles, all-edge HT feasibility, dual-gradient finite differences, analytical metric-gap examples, cache equivalence, and abstention

controls. These tests validate implementation behavior; the mathematical guarantees depend on the stated assumptions, not on test coverage. From the repository root, the following commands regenerate manuscript tables and independently repeat the research protocol:

```
python paper/generate_tables.py
bash scripts/reproduce_research.sh smoke
bash scripts/reproduce_research.sh full
```

The full command writes new results below `results/research_reproduction`, preserving the original measurements. Individual benchmark commands are also available:

```
export PYTHONPATH=src
export OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1
python -m earlyht.v2_benchmark --fast-baselines \
  --groups main sensitivity heldout n12 n16 --repeats 3 \
  --output results/paper_reproduction/v2_fast
python scripts/benchmark_cp_v2.py \
  --output results/paper_reproduction/v2_cp
python scripts/benchmark_ht_v3.py \
  --groups small band band_fresh control \
  --joint-schedule early --baseline shared --repeats 3 \
  --output results/paper_reproduction/v3_early \
  --figures figures/paper_reproduction/v3_early
```

The paper uses the strengthened-baseline records in `results/v2_fast` and the early-schedule HT records in `results/v3_early`. Earlier implementation variants remain in the repository for traceability but are not substituted for those comparisons.

D.6 Compression-gap and excess-error controls

For the diagonal comparison, write $q_i = d_i/m$, where $m = \min_i d_i$ on the retained set. The scalar bound is evaluated by the one-dimensional convex problem

$$\min_{t \geq 1} \left[t^2 + \sum_{q_i > 1} \frac{(\sqrt{q_i} - t)_+^2}{q_i - 1} \right].$$

Its monotone derivative is solved by a bracketed scalar root finder. Independently, the simplex dual in Proposition 5.2 is solved from its KKT multiplier. The computations include both an active boundary $t = 1$ and interior optima, with repeated minima handled separately. All matched subsets are enumerated for $k = 1, \ldots, 10$, while the proven suffix reduction is used for the larger values in Table 8. These calculations operate on diagonal vectors; the large matrix or full PDE solution is not assembled.

The tie control fixes the cumulative center-step budgets to $(0, 2)$, allows at most three fits per stage, performs no ALS sweeps, and disables joint refinement. The optional tolerance is 10^{-8} in absolute squared-energy units for unit-norm forcing and $\mathcal{A} = I$. Each returned tensor is checked against the exact best root-unfolding gain. This control reports work counts, not timing ratios. The source generators directly specify the forcing; their exact solutions are used only as evaluation oracles.

The extension tests cover independent scalar primal/dual calculations, exhaustive small-subset comparison, the actual rank-one product energy, and the returned tensor's excess error. An additional compatibility audit compares the new default strict policy with the archived original source on 22 cases; selected trees, interval values, margins, and work counts agree exactly. Reproduction commands are

```
export PYTHONPATH=src
python scripts/benchmark_compression_gap.py
python scripts/benchmark_excess_energy.py
python -m unittest discover -s tests -v
```

The new records are in `results/compression_gap` and `results/excess_energy`. They do not replace the original timing records. Both benchmark scripts accept alternate output paths.

k	Scalar	Best scalar + tail	Omitted	$k^{2/3}/4$
16	11.60	7.67	3	–
32	25.35	13.12	6	–
64	54.50	22.04	12	4.00
128	114.96	36.64	20	6.35
256	238.71	60.38	35	10.08
512	489.71	98.86	58	16.00
1024	995.95	161.04	97	25.40
2048	2013.40	261.30	160	40.32
4096	4054.05	422.64	261	64.00

Table 8: Optimized scalar relaxation values. The product bound and actual rank-one gain equal one in every row. "Scalar" optimizes the center in the full space; the next column additionally optimizes spectral compression and includes the exact tail. The lower-bound theorem applies to even $k \geq 64$.

E Reproducibility and draft status

The accompanying repository contains complete proofs, source code, deterministic input generators, per-call results, plotting scripts, and unit tests with recorded run logs. The tests cover independently assembled operators, energy identities, feasible rank bounds, spectral-tail enclosures, joint-dual gradients, cache equivalence, and abstention on ties. Tests of known exact solutions are used as implementation oracles; experimental forcing constructions do not use manufactured target solutions. The original numerical tables are generated from recorded result files by `paper/generate_tables.py`. The compression-gap and excess-error tables are generated by their corresponding benchmark scripts. The data and tables are not manually retuned during manuscript preparation. The affine-forcing study is recorded under `results/parametric_reuse_n8` and `results/parametric_reuse_n12`; its complete-call logs include three repetitions, source hashes, all abstentions, and per-query candidate gains. `scripts/report_certificate_reuse.py` checks the saved records and regenerates the additional tables and figure. The benchmark script keeps evaluation-oracle work outside all timed selection and fitting calls and validates pointwise repetition consistency.

From the repository root, a quick end-to-end check is `bash scripts/reproduce_research.sh smoke`. Replacing `smoke` with `full` reruns the main studies. New outputs are written to a separate reproduction directory. The appendix documents which cases informed development and which were held back until the final policy was fixed. All abstentions and slower cases are retained.

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